

Hybridization and artificial intelligence in optimizing university examination timetabling problem: A systematic review

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Abstract

University Examination Timetabling Problem is the most important combinatorial problem to develop a conflict-free timetable to execute all of the exams in and with the limited timeslots and other resources for universities, colleges or schools. It is also an important Nondeterministic Polynomial Time (NP)-hard problem that has no deterministic solution, so it is an important study to develop an optimised solution to satisfy all of the hard and soft constraints required to create a timetable that would be flexible for all stakeholders such as students, teachers, invigilators, as well as management. Several heuristics, meta-heuristics and mixed integer programming approaches have been performed in the past to develop a solution for it, but use of hybrid techniques and hyper-heuristics through implementing Artificial Intelligence (AI) improve the efficiency and performance in solving the problem using the single or multiple objective function. In this systematic review, we are going to learn the effectiveness of these heuristic and meta-heuristics techniques in solving the examination problem as well as how these can be merged to develop a more effective and efficient hybrid solution. The impact of AI is also reviewed in developing the solution for the examination timetabling problem. While analysing papers included in this SLR, various research gaps are identified including the level of hybridisation between population-based

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and local-search techniques, designing an effective algorithm for an initial solution, providing a balanced solution for multiple stakeholders, and developing a phase-wise and partial solution to improve the effectiveness of a final algorithm for the university examination timetabling problem. It provides insight into the various challenges and opportunities as well as future directions to develop or to perform research in providing a solution for it. It is identified that the use of machine-learning, reinforcement learning, and deep-learning with more advanced AI techniques can optimise and to improve the performance of a solution for the university examination timetabling problem.

KEYWORDS

artificial intelligence, examination timetabling problem, hybrid algorithm, university timetabling problem

Context and implications

Rationale of this study: This literature review was conducted to present an efficient and optimised solution for the university examination timetabling problem (UETP) by implementing hybridisation and artificial intelligence (AI) techniques. The objective is to maximise satisfaction for students, teachers, invigilators and management to ensure examination activity in universities without any resource conflict.

Why the new findings matter: The review highlights the positive impact of hybridisation and AI techniques including machine-learning, deep-learning and reinforcement learning by exploring the new dimensions to develop or enhance an algorithm that provides an optimised solution for the university examination timetabling problem with greater efficiency and accuracy.

Implications for researchers and educational institutions: The Implication of this study is significant for universities or educational institutes in providing an efficient and optimised technique to conduct examination activities. It provides a comprehensive guide to minimise all of the conflicts in resource allocation during examination activities, ensuring a satisfactory and optimised solution for all stakeholders including students, teachers, invigilators and management to a greater extent. This research offers an opportunity to researchers to explore the application of hybridisation and AI techniques, including machine-learning, deep-learning and reinforcement learning in developing and enhancing an efficient and optimised algorithm for the university examination timetabling problem.

INTRODUCTION

The University Examination Timetabling Problem (UETP) is the leading combinatorial optimisation and Nondeterministic Polynomial (NP) Time-hard problem that cannot be solved in polynomial time (Aldeeb et al., 2021). The UETP is one of the extensive and large-scale

administrative activities in various business operations where scheduling is required to allocate resources without conflict, especially developing conflict-free timetables to conduct classes and examination activities in schools, colleges and universities (Rahani et al., 2021). It is required to prepare a conflict-free timetable to schedule all of the classes or examinations successfully. Examination is a significant activity conducted during or at the end of a semester. One of the major requirements is to regulate thousands of exams for students of all courses without any error or conflict in the limited time and resources. An optimised timetable is necessary before conducting any semester or examination, which is usually a routine business process in these institutes (Godwin, 2022). Any delay due to invalid slots or unavailability of exam slots being on time will also affect student performance and managerial activities (Ajik et al., 2023). An optimised timetable schedules all of these examinations for particular courses for all students without conflicts. These conflicts include duplicate exam slots, unavailability of room, or overlapping exam duties of invigilators. Several resources such as teachers, invigilators and rooms are allocated to conduct exams by eliminating these constraints to make a flexible and feasible solution. It becomes difficult to arrange such an extensive examination process without a managed and organised examination timetable. An invalid timetable leads to missing exams, duplicate room allocations, duplicate student exams, etc. So, these conflicts lead to an unsuccessful examination process. Universities manage various activities more efficiently by using information technology and efficient algorithms (Williamson, 2018). It is impossible to get a fully optimal solution due to the complexity of the timetabling problem. Therefore, it can simply identify an optimal or near-optimal solution from a list of available solutions (Aldeeb et al., 2021). It involves optimising various constraints related to exams, rooms, students, invigilators and teachers. It tries to satisfy all of these constraints at the maximum level to find the best possible solution. Educational institutes are performing several different activities to execute their business processes. Based on the structure of educational institutes and their business processes, timetabling problems can be classified as either a school timetabling problem or a university timetabling problem. The university timetabling problem is similarly divided into university course timetabling problems (UCTP) or UETP (Bashar & Aldeeb, 2019). The UCTP deals with the successful conducting of classes of all courses during the semester, whereas UETP is responsible for conducting examination activity without conflict. UETP is simply the allocation of resources such as rooms, invigilators and available timeslots to conduct the exams of various courses for different teachers for several students. UETP can be further classified into capacitated and uncapacitated UETP (Ashis Kumar Mandal & Kendall, 2020). In an uncapacitated timetabling problem, there is no room capacity constraint, whereas it is considered a hard constraint in capacitated UETP.

The examination scheduling involves scheduling several exams of various courses for students within a given period of time by allocating available timeslots after optimising all of the constraints. Constraints are the physical limitations of these resources when conducting exams, such as a limited number of rooms (Leite, Fernandes, & Melicio, 2018). During this process, one needs to assign limited timeslots and invigilators without any conflict to thousands of exams for several courses with different teachers.

Different stakeholders, students, teachers, invigilators and management are involved in the examination process. Every stakeholder wants a timetable to accomplish their objective and flexibility in conducting examination activity. Getting an optimised and flexible timetable for all stakeholders is not easy. Therefore, the timetabling problem tries to optimise one or more objectives for these stakeholders (Li et al., 2018). Based on achieving these objectives by optimising the examination timetabling problem, the optimisation approaches can also be classified as single-objective and multi-objective approaches (Ghafar et al., 2021). Single-objective approaches perform an algorithm to optimise a single objective function without any weightage to specific stakeholders. In contrast, multi-objective approaches try

to optimise multiple cost functions to optimise the solution for all stakeholders. It can be possible to develop a solution with multi-objective approaches that could be flexible for one or all stakeholders.

The objective of the university examination timetabling problem is to satisfy hard and soft constraints by minimising the cost function. Constraints are physical limitations of the available resources or limited amount of time to use that resource. An understanding of these constraints is necessary to optimise the university examination timetabling problem. These constraints are classified as hard and soft constraints. Hard constraints must be satisfied to create a feasible solution for the timetabling problem. Otherwise, it cannot be possible to execute all exams for all courses within the specified time. Soft constraints involve providing quality in the generated solution and facilitating a flexible timetabling solution for one or multiple stakeholders (Bashar & Aldeeb, 2019).

A solution for examination scheduling is simply a system that satisfies all of the hard constraints and tries to eliminate soft constraints in such a way that leads to conducting all of the exams for different courses for all students in available timeslots without conflict. There would be no polynomial solution available. Therefore, any solution for UETP tries to minimise the cost function by optimising all of the constraints for all possible solutions (Bashab et al., 2020). Heuristic and meta-heuristics approaches can implement an optimal solution for examination scheduling with maximum performance in a reasonable time. These approaches try to identify a feasible solution to schedule all exams within the given timeslots in the provided rooms by satisfying all hard constraints and minimising the cost of soft constraints (Bashab et al., 2020). A flexible and satisfactory solution minimises multiple cost functions for different stakeholders (Li et al., 2018). A hybrid algorithm that combines various algorithms with AI-based techniques would be the more desirable solution for a university examination timetabling problem (Ashis Kumar Mandal & Kendall, 2020) and it can be identified with the proper use of hybridisation and application of Artificial Intelligence (AI). If a selected exam has a scheduling conflict, it will be moved. Otherwise, it will remain unchanged. Techniques to improve the performance and results of the algorithm. It means there is still a huge gap to improve the performance of the algorithm for a UETP. The level of hybridisation between population-based and local search techniques will provide a more optimised algorithm (Abayomi-Alli et al., 2020). It would be an effective solution to develop a technique to get an optimised and feasible initial solution using Machine Learning (ML) or Deep Learning (DL) techniques (Dewi et al., 2021). In the same way, use of various AI approaches can be more effective to minimise cost functions for different stakeholders and generates a more optimised and efficient algorithm with intelligent techniques and machine-learning (Bashab et al., 2020). AI techniques such as DL find the best solution using historical data and Reinforcement Learning (RL) can optimise the previous solution to find a more optimised and intelligent solution (Pillay & Ozcan, 2017).

In this review paper, the following research questions are formulated for conducting a review of a UETP on the basis of identified research gaps to develop a baseline for new researchers in this field to learn about more advanced algorithms embedded within AI and hybrid techniques.

- Q1. What are various conflicts and constraints used by researchers to optimise UETP?
- Q2. How to implement a solution for UETP to achieve multiple objectives to satisfy students, teachers and management?
- Q3. What are the major techniques and algorithms used to solve university examination timetable problems, and what are the main constructs of these techniques?
- Q4. How do different algorithms combined with AI techniques to create hybrid algorithms improve the efficiency and performance to optimise UETP?

This Systematic literature review (SLR) provides an opportunity for research to develop or improve any solution or algorithm for UETP by implementing it using techniques such as hybridisation and AI. There is no need to focus on a complete UETP solution, as this SLR guides new researchers to optimise a part of the solution to get an optimised overall solution. So, this SLR would be a new milestone to provide new dimensions to implement a UETP to obtain a more optimised algorithm by using hybridisation and AI techniques such as ML, RL and DL.

The rest of the paper is organised as follows: Section 2 describes the main objectives for answering the research questions in this SLR. Section 3 describes the related work and background study in the field of UETP, and Section 4 describes the SLR protocol used in this paper. Then, Section 5 formulates the UETP and describes various constraints in it. Section 6 presents results and summaries of a detailed analysis of all the techniques and algorithms to solve the university examination timetabling problem. Section 7 explains the research gaps, future trends in this research field, ways to improve the performance, accuracy of examination scheduling algorithms, and implementation of various AI techniques. It also suggests future implementation of various approaches along with AI techniques. Section 8 concludes the topic.

AIMS AND OBJECTIVES

The study is to identify and get an overview of all the relevant research in the field of UETP and to learn about different techniques and algorithms to optimise it. The overall purpose of this research is to understand the mechanism to combine various optimisation techniques to create a hybrid algorithm with AI techniques to improve the accuracy and efficiency of the solution. It also tries to understand the UETP and all known conflicts and constraints in optimising university examination timetable problems. Various optimisation techniques and algorithms are analysed to get an efficient solution to the problem. It also tries to identify ways to achieve multiple objectives to satisfy all stakeholders such as students, teachers, invigilators and management. The main focus of the study is the use of hybridisation with AI to optimise the university examination timetabling problem by improving the performance and efficiency of available algorithms.

Significance of this research

The SLR procedure describes the state-of-the-art techniques and algorithms effective in implementing a hybridisation approach with AI techniques to solve a UETP. It provides an opportunity to improve the efficiency and performance of already available solutions. The review explains all of the details concerning the university examination timetabling problem and the constraints to be optimised in it. It also describes in detail the different approaches used in implementing the hybridisation approach followed by various researchers in solving the problem. Based on this review, it explains and discusses all of the research gaps, open issues and potential techniques to build a novel and efficient solution for examination scheduling using hybridisation and AI techniques.

This SLR summarises and identifies all techniques and algorithms used to optimise UETP. It also describes all the constraints to be satisfied to minimise cost functions related to student, teacher, invigilator and management. It also provides a review of the application of various algorithms in terms of their performance and accuracy in optimising the problem. It mainly focuses on the importance of hybridisation and ways to combine different approaches to implement it. It explores numerous AI techniques including ML, DL and RL

to get more robust and intelligent solutions. Finally, this review identifies and explains all open issues, research gaps and potential ways to optimise UETP with hybridisation and AI techniques.

Novelty of and need for this SLR

A timetable is one of the crucial documents for any university to conduct its exams, and creating a manual timetable without any conflict is a much more difficult and complex task (Rahani et al., 2021). An examination is a principal event in universities where it is conducted multiple times as a mid-term, end-term, comprehensive or thesis examination. University departments perform the same task by sharing a few resources like invigilators and rooms. It adds an extra layer of complexity in designing conflict-free timetables. There is not yet an optimal and feasible solution available due to the complexity of the problem as it is an NP-hard problem, so the improvement in the already available solution is still there (Aldeeb et al., 2021). There is a strong need to develop more accurate and performance-oriented algorithms and explore all of the techniques and strategies available to design such algorithms to optimise timetabling problems. This research topic focuses on the need for hybridisation and AI techniques to get the optimised solution for the university timetabling problem. So, it provides a state-of-the-art overview of UETP in such a way that will be a constructive and relevant contribution.

RELATED WORK/BACKGROUND STUDY

Most of the previous reviews discussed capacitated or uncapacitated UETP with various techniques but ignored the impact and importance of AI techniques in developing an optimal and efficient solution. Aldeeb's review (Aldeeb et al., 2021) explains the uncapacitated UETP only, without discussing AI and hybrid solutions. Uncapacitated UETP is not a natural problem as room capacity is an essential constraint in implementation in most real-life scenarios. On the other hand, Bashab et al. (2020) follows both course and examination timetabling with meta-heuristics approaches. So, its review would not be a valid choice if someone wants to extend or design in the application of AI and hybridisation. In short, these reviews mostly ignore the use of AI techniques and hybridisation to implement an examination timetabling problem. So, researchers cannot identify a clear-cut approach to find a more efficient and optimised solution to solve a timetabling problem.

The proposed systematic review will be a good contribution for researchers to implement the advancement of AI and ML techniques with hybridisation in finding the optimal solution for UETP. It provides a multidimensional facet to analyse the capacitated and uncapacitated UETP by implementing AI and hybridisation. It is a good contribution for researchers to implement the advancement of AI and ML techniques with hybridisation in finding an optimal and robust solution for examination scheduling. It also critically analyses various algorithms and approaches to identify their pros and cons while implementing AI and hybridisation.

There are several approaches to solving the UETP, including heuristic, meta-heuristic, hyper-heuristic, evolutionary algorithms and hybridisation of these approaches. AI, ML, DL or RL techniques are embedded to improve the performance and accuracy of these approaches.

Most of the earliest techniques to solve the examination problem are heuristic approaches such as graph colouring technique (Sun & Wu, 2023), constraint programming and integer programming (IP) (Bellio et al., 2021). These mathematical techniques are easy to implement but slow due to complex mathematical calculations. So, they are not able to provide the

best optimal solution. Meta-heuristic approaches can solve the timetabling problem more intelligently than the heuristic approaches (Carlsson et al., 2023). These approaches are local search approaches and population-based approaches. Local search techniques start from a single solution to refine it by removing all of the conflicts by applying the heuristics, whereas later generate multiple solutions and try to find the optimal solution from that group of solutions (Bashab et al., 2020). Population-based approaches will return optimal solutions in less time but not as refined as the local search approach. Hyper-heuristics optimise the problem by generating low-level heuristics during the initial phases of development of the population as well as during the process of refining the solution iteratively (Dewi et al., 2021). These approaches optimise single or multiple objectives by minimising different cost functions. AI has a prime role in implementing all processes, such as examination, lecture delivery, student learning, etc. (Crompton & Burke, 2023). So, implementing these approaches with AI and hybridisation produce an efficient and intelligent algorithm for examination scheduling nowadays (Staddon, 2020). Hybridisation overcomes inherited problems in various optimisation techniques by merging them. AI and ML enhance the performance and accuracy of the algorithm to generate a more robust and intelligent solution for the university examination timetabling problem (Dewi et al., 2021).

Several benchmark and public datasets are available to design an optimal solution for examination scheduling. The International Timetabling Competition (ITC) and Toronto datasets are famous for optimising a solution for UETP (Leite, Fernandes, & Melicio, 2018; Leite, Melicio, & Rosa, 2019). Several researches are implemented with custom datasets to find the optimal solution for a particular situation.

The ITC, organised by several universities and organisations, is a widely known dataset to find an optimised solution for UETP (Li et al., 2018). It presents the real-time situation of university examination timetabling and contains information about eight different instances of exams with various conflicts to be optimised. Several dataset files are available in ITC, in which ITC2007 and ITC2018 datasets are well-known and widely used by researchers (Ceschia et al., 2023).

Toronto datasets, consisting of different instances of university examination timetables, were developed by the University of Toronto (Qu et al., 2009). It includes information about timeslots, rooms and exams with hard and soft constraints related to 13 instances of uncapacitated UETP (Bashar & Aldeeb, 2019).

This paper presents a complete review of both versions of UETP and provides a comprehensive description of the available solutions to optimise them. It also considers issues related to single-objective and multi-objective cost functions for flexible examination timetables for universities. So, this research will be a valuable addition for researchers who want to improve the algorithms by applying hybridisation as well as AI techniques. They want to learn various methods to develop a flexible and generalised solution for the UETP.

RESEARCH METHOD

This paper presents a systematic literature review (SLR) to identify all techniques and algorithms to optimise university examination timetabling solutions. This review is valuable for all researchers who want to study timetabling problems and find an optimised and efficient solution by implementing hybridisation and AI techniques. They can evaluate all of the merits and demerits of these approaches to extend or redesign any previous solution to make it more efficient in terms of time and cost by using advanced and efficient logic. It also critically evaluates those techniques to implement them in a hybrid solution with AI techniques. This review, used in UETP by implementing hybridisation with AI

techniques, primarily focuses on the efficiency and enhancement of optimisation techniques. Researchers could evaluate optimisation techniques and algorithms and learn about the impact of hybridisation with AI-based techniques. This review could be a starting point to design a new algorithm to solve the problem or extend previously available algorithms to optimise the UETP. For this purpose, we performed searches for this SLR in different online databases and repositories as shown in [Table 2](#), such as ACM, Elsevier, IEEE and Springer, using multiple terms such as ‘examination timetabling’ and ‘examination timetable’ in the titles of papers, and then to refine the searches’ number of keywords such as ‘university examination timetabling problem’, ‘examination timetable problem’, ‘population-based techniques’, ‘local search techniques’ and ‘hybrid techniques’. It also identifies from the searches that no proper review focuses on optimising the UETP with hybridisation with AI techniques.

Systematic literature review

The strategy as described in Masrom et al. (2021) is followed in this review paper by planning in the first phase, then conducting the review in the second phase, and generating results and summaries in the third phase, as shown in [Figure 1](#). The review method proposed in Bashab et al. (2020) also applied a few modifications in the search strategy to guide the researcher about the use of hybridisation with AI techniques in the timetabling problem. In addition to this search strategy, the backward snowballing technique (Wohlin, 2014) was also applied to include important papers related to our research topic but skipped due to the previous SLR strategy.

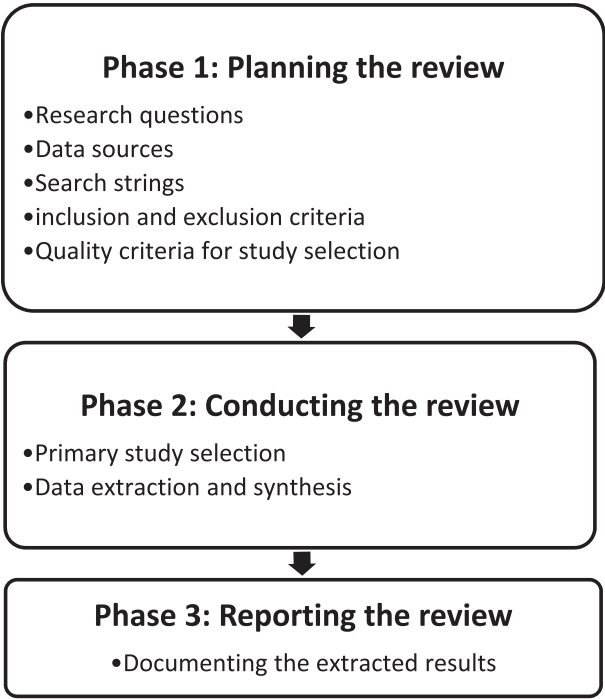


FIGURE 1 Phase-wise workflow of systematic literature review process used in this study.

Phase 1: Planning review

Research questions

The results questions finalised for this research are presented in Section 2 already. These research questions are to achieve the objectives defined in this SLR.

Data sources

Digital Sources selected for current research are

- IEEE
- SPRINGER
- ELSEVIER
- ACM

There are some other multidisciplinary databases such as Scopus and Web of Science, but these are considered as secondary databases and include research papers and abstracts that are already indexed in the selected databases above. Web of Science mainly focused on pure sciences, medicines, social sciences and Art & Humanities as compared to the selected databases (InCites Benchmarking & Analytics, 2025), whereas this SLR is related to the computer science domain to design an optimisation algorithm using hybrid and AI techniques.

Search string

The most important part of the literature review is to build the search query to find the matching research papers related to our topic. It is based on the guidelines in Bashab et al. (2020) and is modified according to the research questions in this paper. First, we have identified the keywords in our research questions as in Section 2. After that these keywords are adjusted with a few related synonyms such as timetable and timetabling. As the domain of this research is a UETP, we have included it to construct keyword phrases to find papers against their titles. The term 'examination scheduling' is also used in most of the papers but 'examination timetabling' is more commonly available alongside it. Therefore, the 'examination scheduling' keyword is not included in the query search string. Then we identify the considerable techniques in developing hybrid algorithms such as 'population-based', 'local search solution' and 'hybrid solution' (A2). The main keywords such as 'examination timetable' or 'examination timetabling' (A1) are searched from paper titles, whereas other keywords are searched from abstracts and keywords. Various synonyms or alternative phrases are arranged with the 'OR' operator to construct the main search query. The final search queries to find the relevant papers for review in this SLR are shown in Table 1.

Inclusion and exclusion criteria

The above filtering process returns the number of papers, including journal, conference or thesis papers as well as implementations of UETP in irrelevant domains. To set the scope of the study relevant to the UETP, it is required to select papers that fulfilled certain requirements and objectives set for this research. It also includes some rules to select the quality publications based on certain criteria, such as only HEC-recognised journal papers to develop an authentic and valid review for UETP research that fulfil the objectives of this SLR. The purpose of inclusion and exclusion criteria is to retrieve only the relevant publications to strictly fulfil the objectives of this review and a focused guideline for the future researchers to develop an algorithm for an effective solution for UETP. For this purpose, the following set of inclusion and exclusion criteria are constructed for this research.

TABLE 1 Search string used in SLR.

Keywords	Search from	Query search string
University Examination Timetabling (A1)	Paper Title	'examination timetable' OR 'examination timetabling'
University Examination Timetabling (A1)	Paper Abstract and Keywords	'examination timetable problem' OR 'university examination timetabling problem' OR 'examination timetable' OR 'examination timetabling' OR 'timetable problem' OR 'timetabling problem'
Hybrid, Local Search, Population-based Algorithms (A2)	Paper Abstract and Keywords	((('hybrid genetic' OR 'local search' OR 'population based') AND ('algorithm' OR 'solution')) AND ('examination timetabling' OR 'examination timetable'))

- IC1: Publication must cover the UETP related to universities, colleges or schools.
- IC2: The research paper must cover the implementation of different techniques, such as heuristics, meta-heuristics or hyper-heuristics to solve UETP.
- IC3: The paper must include the strategy to construct, improve or extend a few techniques to construct a hybrid solution with AI to solve UETP.
- IC4: Paper must be published in a recognised journal of X, Y, or W category only.
- EC1: The paper describes the timetabling problem in other domains instead of educational institutes.
- EC2: Paper that is a letter to the editor, abstract, thesis, or published in a conference.
- EC3: Paper is published in a language other than English.

Phase 02: Conducting review

Primary study selection

After constructing the search query, we executed this query with four different electronic repositories – ACM Digital Library, IEEE, Elsevier and SpringerLink – on 16 May 2023. The publication limit between 2018 and 2023 inclusive is set to limit the search. These electronic databases are chosen due to their importance in publishing articles related to hybrid and AI-based algorithm development for optimisation problems. The search is performed on the title, abstract and keywords. The first search query is performed on the title only, and the second is performed on abstracts and keywords. The results of these search queries against each selected database are given in Table 2, that shows the number of research papers published, excluding books, chapters and conference papers. There are 51 Scopus indexed papers that are already available in the selected database, so it will be a good reason to exclude Scopus from the list.

Data extraction and synthesis

The Tollgate approach (Afzal et al., 2009) is adopted for the paper filtration process in this SLR. There are 215 research papers after performing the above query strings. A critical screening process is performed to identify the exact paper to review. Figure 2 shows the filtered papers after each iteration using the Tollgate approach. For simplicity, we have used P1 (papers after initial query), P2 (papers left after removing duplicate), P3 (application of EC2), P4 (application of EC1 and IC1), P5 (application of IC4) and P6 (after EC2 and EC3) for each iteration. In the first phase of this screening process, we left 151 papers after removing 64 duplicate papers. In the next step, all conference papers are removed using EC2, and we have 115 journal papers in our list. Few papers are excluded based

TABLE 2 Search query result of selected databases.

Database	Results (papers only)
Elsevier	23
ACM	10
SpringerLink	40
IEEE	31

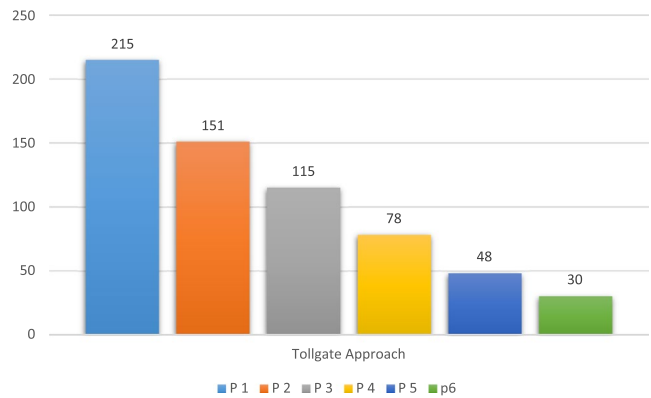


FIGURE 2 Paper categories for selected papers based on filtering criteria using the Tollgate Approach in the SLR process.

on the title by applying IC1 and EC1 to select only those related to UETP and that left 78 papers. We left 30 after applying IC4 to have only HEC-recognised X, Y or W category journal papers. The next step of screening is to download all of these papers and remove non-English papers by applying C3. Now the remaining 30 papers are selected for detailed investigation by reading their abstracts, results and conclusions in the third phase of the screening process. The second two inclusion criteria are critically applied here to get papers with the application of any of the techniques to optimise UETP with hybridisation and AI techniques. All papers fulfilled the requirements of EC2 and EC3, so they were selected for the final review of the UETP.

To support the concept of selected papers, we have applied the backward snowballing technique (Wohlin, 2014) to find any potential papers by analysing their references included in the literature review of related work. We have identified six unique papers with alternative keywords, which are not covered by our inclusion and extrusion criteria and also ignored the already selected papers. Duplicate papers are ignored during this process. These papers must be reviewed to identify the techniques applying hybridisation and AI techniques in solving the UETP. In the second iteration of backward snowballing, no further paper fulfilled our inclusion and exclusion criteria. Thus, we have six new papers for review after performing the backward-snowballing approach.

Exceptional papers

In our screening process, we skipped many papers not published in the peer-reviewed scientific journals of categories X, Y or Z. However, those are well-defined and cited several times by other researchers for their study of UETP and have extensive implementations of different hybridisation and AI techniques for it. To avoid this bias, we have selected two such papers for full-text reading to include in our final review. We therefore have 38 (30 + 6 + 2) papers for final review at the end of this SLR. Out of these selected papers, 30 papers are

selected using the Tollgate approach as shown in Figure 2. For time reference, Table 3 shows the distribution of selected papers with different publication years.

Data filtering and labeling

After paper selection for review, these are categorised by techniques used to solve the UETP. These papers are labelled based on the algorithm developed in it. A few of the papers are included in multiple categories. All of these classifications are double-checked to avoid any ambiguity for final review. Thus, we analysed five different approaches and 19 different algorithms used to get a hybrid solution and follow AI techniques shown in Tables 6 and 7. In conducting the initial review of the papers in this SLR, we have categorised these papers in various classes. Table 4 describes the distribution of these papers in this SLR to describe the UETP as capacitated, un-capacitated UETP, or both. Several researchers follow the single-objective or multi-objective approaches in optimising the UETP shown in Table 4, and the same also shows the use of various benchmark datasets to optimise university examination timetabling in the selected papers for this SLR. It further explains some objective functions in the case of a multi-objective approach. Table 5 describes the constraints implemented to optimise the UETP in these papers.

Phase 03: Reporting the results

In the third phase of the SLR, the results of the review after analysis are generated in Section 4, giving the complete description of all of the techniques and algorithms used to optimise the UETP, and how they can be mixed to achieve hybridisation. It also describes the impact of AI on these algorithms and techniques in terms of performance and efficiency. So, it could be a prime gateway to begin any new study in this field.

TABLE 3 Paper distribution year-wise.

Year	No of papers
2018	4
2019	11
2000	7
2021	6
2022	5
2023	4

TABLE 4 Summary features of selected papers.

Feature	No of papers
Capacitated UETP	25
Un-Capacitated UETP	8
Both Capacitated and Un-Capacitated UETP	4
Single-objective UETP	32
Multi-objective UETP	6
ITC Dataset	6
Toronto Dataset	14
Custom Dataset	21

TABLE 5 List of hard and soft constraints for various stakeholders.

Stakeholder	Hard constraints	Soft constraints
Student	• All of the exams must be conducted only for registered students • Avoid multiple exams for student in same slot • All of the exams for students must be conducted within a specified time • Specific exam ordering must be observed	• No two exams in a day or consecutive exams for student • In split exams there must be a minimum distance between rooms • There would be no more than x exams in a day for a student
Invigilator	• No duplicate duty in the same slot • Invigilator and teacher must not be the same • There would be no exam without an invigilator	• There will be no more than x exams for an invigilator • Only x exams can be assigned to an invigilator in a day • No consecutive duty for an invigilator • No split exam for one invigilator
Teacher	• No multiple exams for teacher in same slot • Invigilator and teacher are not the same • A few exams must not be simultaneous, and a specific room/room type is required for a particular exam • Must be optimised for a feasible solution	• Teacher, expert, supervisor or invigilator must be same • Exam is conducted in single room, fixed timeslot is required for a particular exam • No consecutive exam for teacher • There would be no x exam in a day for a teacher
Management	• All exams must be scheduled within a specified time, exam length is less than or equal to room capacity • Exam conducted within fixed timeslot • All of the resources must be available for each slot • Shared exams have fixed time slot for each department	• No more than x exams in a day, provision for empty slot or Friday prayer slot, no split exam in multiple rooms, avoid exams in same room, large exam must be conducted early, and allocating multiple timeslots for an exam

RESULTS AND ANALYSIS

The results of the proposed SLR conducted in this paper are reported based on RQ1 to RQ2. In this section, we will map our results with the research questions mentioned above in detail.

University examination timetabling problem (UETP) and its associated constraints via SLR (RQ1)

The purpose of RQ1 is to formulate the UETP and to identify various constraints implemented to optimise the timetabling problem.

University examination timetabling problem formulation

This section describes the basics of UETP construction which are required to formulate it before implementing any optimisation algorithm. The UETP can be defined formally as a set of exams e , rooms r , invigilators i and timeslots n as in [Equations \(1–5\)](#), where all of the e exams for all students must be scheduled in available rooms in set r during the available timeslots n by teachers t (Ashis Kumar Mandal & Kendall, [2020](#)). The objective

of the university examination timetabling problem is to minimise the *obj* function, given in Equation (7), with multiple objectives achieved by satisfying a set of soft and hard constraints *c*, as given in Equation (6) (Ghafar et al., 2021).

$$e = \{e_1, e_2, e_3, \dots\} \quad (1)$$

$$R = \{r_1, r_2, r_3, \dots\} \quad (2)$$

$$n = \{n_1, n_2, n_3, \dots\} \quad (3)$$

$$i = \{i_1, i_2, i_3, \dots\} \quad (4)$$

$$t = \{t_1, t_2, t_3, \dots\} \quad (5)$$

$$c = \{h_1, h_2, h_3, \dots, ss_1, ss_2, ss_3, \dots, si_1, si_2, si_3, \dots, st_1, st_2, st_3, \dots, sm_1, sm_2, sm_3, \dots\} \quad (6)$$

$$obj = \{f_s, f_i, f_t, f_m\} \quad (7)$$

Representation of timetabling problem

Before discussing the techniques to solve the examination scheduling, it is very important to discuss how to present and formulate this problem. It can be presented with different approaches in order to solve the UETP by identifying various conflicts easily. Using graph colouring and 'symmetric bit' are main techniques for this purpose (Bashar & Aldeeb, 2019)

Graph Colouring

To formalise the UETP, several researchers followed the graph colouring problem that was implemented by Neufeld and Tartar, (1974). Exams are represented using nodes, whereas edges represent conflicts in the graph as Equation (8). Node colours represent timeslots to conduct a particular exam. Adjacent nodes having the same colour represent conflicts, so the main objective is to minimise the adjacent nodes having the same colour.

$$G = (N, E) \quad (8)$$

N is represented as the total nodes or exams to be scheduled, and *E* represents the edges or conflict between two nodes or exams.

Symmetric bit matrix

The timetabling problem is represented with a bit matrix of *N* × *N* having *N* number of exams that shows several conflicts of exam *i* with exam *j*, otherwise 0, as shown in the matrix given below (Cole, 1964). It may be one or many conflicts between two exams. It can be easily converted into a colour graph and vice versa. For example, in the given matrix, exam 1 has

one conflict with exam 2, and two conflicts are associated with exam 4, whereas there is no conflict with exam 3.

Symmetric bit matrix				
	1	2	3	4
1	–	1	0	2
2	1	–	1	0
3	0	1	–	0
4	2	0	0	–

Constraints in the UETP

This section describes all the constraints that are required to avoid all conflicts in university examination timetabling to generate the optimised timetable. These constraints are divided into hard and soft constraints for various stakeholders, as shown in Table 5. These constraints depend upon the situation where they are implemented, but most are common for every UETP (Bashab et al., 2020). However, it is necessary to resolve all constraints to the maximum extent to get a global solution to the UETP. There will be no acceptable and feasible solution if any hard constraint is not satisfied. Soft constraints are not strictly required to resolve the UETP but are necessary to get a flexible solution (Bashab et al., 2020; Godwin, 2022). The available timeslot is the foremost constraint to locate all exams to optimise the UETP.

These constraints are also classified as primary and secondary constraints to satisfy the needs of particular stakeholders to minimise all cost-objective functions (Bashab et al., 2020). Primary constraints are essential to get a minimum optimal level of UETP. They must be implemented to generate a feasible solution for the problem. On the contrary, secondary constraints are required to optimise a timetabling solution for a particular domain. They are used to develop a generic solution for examination scheduling to create a conflict-free timetable for universities. ITC and Toronto datasets contain many instances related to hard and soft constraints, which are essential for developing an optimised solution. The benchmark results of these databases in satisfying these constraints are available in Ceschia et al. (2023).

UETP solution based on objective functions via SLR (RQ2)

This section describes objective functions in implementing the UETP to answer RQ2. The objective function is to optimise the UETP to achieve some specific objective or satisfy some specific stakeholder (Ashis Kumar Mandal & Kendall, 2020). So, the maximum number of objective functions used in optimising the examination scheduling depends upon the number of stakeholders involved in getting any benefit from the solution. There are four stakeholders – student, teacher, invigilator and management – to minimise the cost objective function (Bashar & Aldeeb, 2019). If the purpose of optimising examination scheduling is to minimise the cost of every objective to satisfy every stakeholder, then it is called a multi-objective UETP. A cumulative cost function integrates all individual objective functions using Equations (10–13) (Lobo et al., 2000) to find the overall optimised UETP. However, a single-objective UETP tries to minimise the single-objective function irrespective of the number of stakeholders (Ghafari et al., 2021).

Objective functions in UETP

After review of the research papers, it is found that the following objective functions must be satisfied to optimise multi-objective UETP. These objective functions are developed to satisfy the stakeholders of UETP such as students, teachers, invigilators and management. For a single objective solution – a cumulative cost function as given below – is satisfied for all of the constraints irrespective of any stakeholder in optimising the UETP (Bashar & Aldeeb, 2019).

$$f(x) = \sum_{t=1}^{\infty} \left(\frac{1}{\left(1 + \sum_{c=1}^{\infty} (wC_h + wC_{ss} + wC_{si} + wC_{st} + wC_{sm}) \right)} \right) \quad (9)$$

Objective function for student

A student is the most prominent stakeholder in the timetabling problem and also requires more flexibility as compared with other stakeholders (Ghafar et al., 2021; Sun & Wu, 2023). He is the person who has to take some exams for different courses within the specified time. An objective function as given below in Equation (10) will minimise the cost value of all constraints to implement a solution for the student.

$$f_s(x) = \sum_{t=1}^{\infty} \left(\frac{1}{\left(1 + \sum_{c=1}^{\infty} wC_{ss} \right)} \right) \quad (10)$$

Objective function for invigilator

An invigilator is another important stakeholder in the timetabling problem and has to observe several students for a particular course. It is also a prime objective to optimise the university examination timetable to provide flexibility for an invigilator to perform examination activity effectively. An objective function for an invigilator, as given below in Equation (11), is to minimise the cost value of all of the constraints to implement the solution for the invigilator.

$$f_i(x) = \sum_{t=1}^{\infty} \left(\frac{1}{\left(1 + \sum_{c=1}^{\infty} wC_{si} \right)} \right) \quad (11)$$

Objective function for teacher

A teacher is a stakeholder who has to prepare exams for some courses for many students. He also needs flexibility to some extent while conducting multiple exams. It becomes complex to handle various course exams at the same time. An objective function for teachers given below in Equation (12) (Ghafar et al., 2021) is to minimise the cost value of all constraints to implement timetabling solutions.

$$f_t(x) = \sum_{t=1}^{\infty} \left(\frac{1}{\left(1 + \sum_{c=1}^{\infty} wC_{st} \right)} \right) \quad (12)$$

Objective function for management

The management is a stakeholder who has to arrange all physical resources like rooms, timeslots and equipment to conduct various course exams for all students. Flexibility is not a prominent issue for management, but efficient utilisation of resources within a minimum time is needed. An objective function for management, as given below, is to minimise the cost value of all constraints to develop an optimised solution for management (Ghafar et al., 2021).

$$f_m(x) = \sum_{t=1}^{\infty} \left(\frac{1}{\left(1 + \sum_{c=1}^{\infty} wC_{sm} \right)} \right) \quad (13)$$

The overall fitness of these multiple objectives is calculated by the formula given below (Lobo et al., 2000). It is simply the sum of all objective functions with objective functions to minimise hard constraints.

$$f_o(x) = \sum_{t=1}^{\infty} \left(\frac{1}{\left(1 + \sum_{c=1}^{\infty} wC_h \right)} \right) + f_s(x) + f_l(x) + f_t(x) + f_m(x) \quad (14)$$

Techniques and algorithms to solve UETP via SLR (RQ3)

It is the prime research question and explains all the details regarding various techniques and algorithms to optimise the UETP. These approaches have been followed several times in past research. Table 6 shows the number of papers selected in this SLR which follow the techniques to solve the UETP, whereas Table 7 provides the paper distribution by the algorithms. Table 4 describes the features of various algorithms used by these researchers in solving the UETP.

Heuristic approaches

Heuristic techniques are problem-solving method to find a satisfactory solution for an optimisation problem like UETP. These techniques are simple and easy to implement but inefficient in getting the best optimal solution (Bashar & Aldeeb, 2019). Graph Colouring technique is the famous heuristics approach to resolve the conflicts in UETP.

TABLE 6 Distribution of selected papers by techniques to solve UETP.

Techniques	Publications
Hybrid Approach	11
Meta-heuristics	14
Heuristics	15
Hyper-heuristics	2
Artificial Intelligence	2

TABLE 7 Distribution of selected papers by algorithms used to solve UETP.

Algorithms	Publications
Memetic Algorithm	2
Simulated Annealing Algorithm	4
Mixed Integer Programming	13
Hill Climbing Algorithm	3
Great Deluge Algorithm	2
Ant Colony Algorithm	1
Firefly Algorithm	2
Graph Colouring Theory	2
Constructive Heuristic Approach	1
Late Acceptance Hill Climbing	1
FruitFly Algorithm	1
Genetic Algorithm	3
Variable Neighbourhood Search	1
Intelligent Water Drop	1
Pre-predator Algorithm	1
Tabu Search	3
Black Hole Algorithm	1
HS Algorithm	1
Particle Swarm Algorithm	1
Artificial Intelligence	2

Graph theory using graph colouring problem

Graph colouring is one the oldest heuristic techniques to solve UETP and is still preferred by many researchers. It creates a conflict-free schedule by allocating timeslots to all exams of various students (Bellio et al., 2021). Exams are arranged first with a graph colouring technique. Then conflicts are solved by using ordering strategies based on heuristics such as largest degree, largest weighted degree, largest enrolment, largest enrolment degree, saturation degree, etc. A minimum number of colours represents minimum conflicts with scheduling timeslots for all exams (Neufeld & Tartar, 1974). It is simple to implement but inefficient to identify an optimal solution. Its performance depends upon the heuristics followed during the constructive method. Due to the complexity of soft constraints, graph colouring is not better at resolving soft constraints (Bashar & Aldeeb, 2019). Instead, it is still famous for developing initial solutions in Al-Betar (2021) and Tilahun (2019) by using colour heuristics in hybrid approaches (Bellio et al., 2021; Leite, Fernandes, & Melicio, 2018) with other techniques to achieve better performance.

Oke et al. (2023) performed a graph colouring approach to schedule the university examination timetable for the Faculty of Science of Federal University Lokoja and successfully implemented all of the exams of 13 departments in three exam slots per day without any conflict. It shows the power of the graph colouring theory in implementing scheduling and optimisation problems.

Construction heuristics

It is a process of solving the UETP step by step. Various algorithms can be used to construct the initial solution, like random assignment, incremental assignment or greedy approach (Bashar & Aldeeb, 2019). In the first step, a blank timetable is initiated, then iteratively assign

exams to different timeslots while satisfying all constraints and maintaining minimum conflicts. After each change, the feasibility of the schedule is confirmed by checking the conflicts between exams. Repeat the process to refine the threshold until a feasible solution without conflict is achieved (Li et al., 2018; Pillay & Banzhaf, 2009). It is simple and easy to implement but cannot always return an optimal solution.

Alkallak et al. (2022) implemented a heuristic approach by implementing the Sudoku grid to allocate courses to exam slots and rooms and successfully remove two exams per student in a day constraint. They also compared it with a simple heuristic algorithm and found that their intelligent algorithm performed better than a classical method in terms of computational cost.

Integer programming

It is a mathematical optimisation strategy to solve the UETP by satisfying the set of constraints to minimise the objective function (Bashar & Aldeeb, 2019). The UETP is a model using a discrete set of integer values to perform discrete decisions to schedule exams for all students within specified timeslots (Sharma & Sharma, 2020). The objective is to reduce all conflicts while scheduling all exams. It works efficiently with a small-sized search space, but it becomes complicated to find optimal solutions with the increased size and complexity. Keskin et al. (2018) models the UETP for Pamukkale University of Turkey using mixed integer programming (MIP) to allocate a single exam to a classroom within a specified timeslot. He defines multiple discrete variables against hard and soft constraints and implements the Two-Stage Heuristics method to optimise the problem in two stages. In the first stage, they develop a solution with slots in a day for all exams, and then assign classrooms and allocate students to classrooms. He tries to minimise multiple objective functions by minimising assigned classrooms, minimising exams in late slots, and trying to minimise soft constraints. He shows his solution performs better than the commercial solution available if the room capacity is greater and provides on average 3.8 conflicts with car92 dataset, but it does not show the same results as the ITC2007 dataset. His approach works well with a small-sized search space, but it is unable to cope with a large-sized search space due to the complexity of the binary and discrete variables used in IP. Another problem with his approach is the low improvement rate, as there is no significant improvement by increasing the execution time.

IP is implemented to formulate examination scheduling in Godwin (2022). They assign many hard and non-linear soft constraints into binary variables to minimise the objective function for student, teacher and management for b-school datasets for Business Schools in India. The proposed solution tries to minimise the cost of multiple objective functions for students, teachers and management by solving several mathematical equations. Multiple objective functions are surrogated to return the feasibility of the overall solution by performing some heuristics to reduce computational costs. It shows 4028 and 0.016 seconds as solving the student and faculty objective function, respectively. In this way individual objective functions were processed quickly as compared to a single objective function. The proposed solution performs a quick and reliable solution with less computation cost. However, it shows reduced performance due to processing a log of mathematical calculations with a large-size search space.

Guler et al. (2021) performs MIP to formulate the UETP of the Industrial Engineering Department of the Yildiz Technical University in Turkey using spreadsheet modelling with Solver Studio Python by implementing constraints related to the department, instructors and students in the form of mathematical equations and tries to minimise the objective function. It is easy to implement but takes time to create a near-to-optimal solution. He also implements a decomposed heuristic approach with MIP to find quick and robust solutions for UETP. It takes less than two minutes to find the solution with a local dataset as compared to a simple MIP that takes much longer.

Mustafa et al. (2022) suggests MIP with various heuristics to create an efficient and robust solution for the invigilator assignment problem or UETP for eight universities in Turkey. His algorithm improves the error rates in manually created timetables by 86%.

A phase-wised solution for UETP is suggested in Al-Hawari et al. (2020) to satisfy hard constraints by using integer linear programming with a benchmark Toronto dataset for the German Jordanian University. It returns an optimised solution quickly with fewer computation requirements by using multiple phases to optimise constraints. In the first phase, exams are assigned to timeslots to avoid adjacent exams in the same slot if taken by the same students. In the second phase, days are allocated to eliminate more than two exams in a day for a student. Rooms are assigned to exams to avoid the room capacity constraint in the third phase. The proposed solution performs better with both datasets in satisfying hard constraints by handling them in multiple phases to balance the computation load. It shows 2.2 students having two exams in a day with Toronto-E92 dataset. However, it is limited to optimising hard constraints only.

Arbaoui et al. (2019) proposed a compact formulation of the timetabling problem to re-solve hard and soft constraints in the ITC2007 and Yeditepe datasets by implementing mathematical equations using MIP. It tries to improve lower bounds for soft constraints like two-in-a-row and two-in-a-day to minimise the single objective function. It shows reasonable reduction of cost of conflicts and drives 29 and 56 cost with yue2013 and yue2023 respectively for a single exam in a day against simple MIP. The proposed system performs better with soft constraints by setting a lower bound with the compact mathematical model, but it becomes difficult to work with large-size datasets.

Rahani et al. (2021) performs a graph colouring technique to optimise the examination timetabling problem with the UiTM Cawangan Kelantan Machang campus dataset. Different hard and soft constraints are presented with colour nodes to formulate the timetabling problem with graph colouring. It starts with the highest-degree to optimise the subjects. It tries to maintain minimum graph colours while connecting exams with student subjects. Then, it implements an objective function to minimise graph colours without the same colour adjacent nodes. It is easy to implement but not efficient in getting a feasible solution. All exams of seven subjects are completed within eight days with two periods using three classrooms having a maximum capacity of 60 students.

The UETP is implemented in Rosyidi et al. (2019) by using IP for Universities Sebelas Maret Surakarta, Indonesia, by developing mathematical equations for various constraints to minimise multi-objective functions. It gives optimised objective function with 29.6 assignments per lecturer as compare to 16.8 assignments per lecturer using manual timetabling.

Exam scheduling was performed by Alrawashdeh et al. (2023) at Al-Hussaein Bin Talal University in Jordan by grouping exams with similar requirements to be scheduled in the same slot to improve the efficiency of the profile-based algorithm using constraint programming. In the first step, all of the exams are grouped with similar characteristics, then the proposed algorithm was used to schedule all exams in available slots, and finally allocate rooms for those exam slots. They found their algorithm performs better in improving the upper bound by student allocation first, so it returns a quality timetabling solution compared to others. It shows a maximum 0.124% conflict of having two exams for a student in a day.

Kasm et al. (2019) suggested integer programming with a time window for Masdar Institute to solve medium and large-size problems with all of the known constraints defined by graph colouring heuristics by allocating a maximum of two exams for a student in a day and found that solution with less computational time. He also revealed that adding more constraints with integer programming increased the algorithm complexity. Solution results are not suitable with most of the publicly available datasets but it takes 0.11 seconds to execute with local dataset Masdar_sem_M4 dataset as compare to 26.33 seconds with simple integer programming.

Meta-heuristic techniques

Meta-heuristic algorithms are the most common techniques to solve combinational optimisation problems with a large-size solution space. It is a process of finding an optimal solution to minimise an objective function for a particular problem by starting with an initial solution that is generated by heuristics techniques. It then decides about the best candidate solution after evaluating iteratively. After a fixed number of iterations, it will return the best possible optimised solution (Bashab et al., 2020). The system reaches a convergence state where the final solution is a near optimal solution. It can solve any problem deterministically, if the calculation to determine the fitness of the candidate solution is not complex. The effectiveness of meta-heuristic approaches is determined by the quality of the initial solution (Bashab et al., 2020). These techniques are classified into single-solution-based and population-based solutions.

Local search-based approaches

Local search-based or trajectory-based techniques perform an iterative process to improve the initial heuristic-based solution by performing a search through neighbourhood structures (Bashar & Aldeeb, 2019). It performs local changes for some fixed iterations to find another solution, which is more optimal than the previous one. It can efficiently find a good solution by exploring it deeply from a large solution space. It is also flexible in performing customised soft constraints that will not be possible in heuristic approaches. However, it is stuck in local optima where it cannot improve the solution by any change (Bashar & Aldeeb, 2019). This problem can be controlled with different local optimisation strategies such as Simulated Annealing (SA), Hill-Climbing (HC) and Tabu Search (TS).

Hill-climbing optimisation

HC optimisation is one of the oldest local search approaches which explores a limited search space to find its optimal value. It starts with an initial solution and iteratively checks part of the solution space to generate a new solution. If there is some improvement in a new solution, it replaces the old one (Al-Betar, 2021). One of the major problems with this technique is being trapped in local optima, but it can be resolved by hybridising with other approaches. B-hill climbing is a modified form of the HC algorithm to solve the UETP by performing the B-operator and N-operator. It performs an initial feasible solution using the saturation degree heuristic, and then iteratively performs the N-operator and B-operator until the convergence or threshold level is achieved. N-operator perform three neighbourhood structures, such as single-move, swap-timeslot and kempe-chain, to move the initial solution to its more fitted neighbourhood solution by selecting random exams. B-operator replaces an exam with a saturation degree heuristic if it is a feasible solution.

It performs the algorithm with Carter uncapacitated datasets and found its performance is better than other heuristic algorithms. The late acceptance HC algorithm is a slightly modified version of conventional HC which improves the local search by avoiding the local optima. It is applied by Al-Betar (2021) and shows its performance much better. It shows the best 157.07 conflicts with STA-F-82-1 dataset, where 4.2 with a CAR-F-92-1 dataset that is near the best previous results.

Tabu search

TS was proposed by Glover and performed its working on the search space by maintaining a tabu list with last visited solutions. It avoids duplicating searches on already visited solutions to avoid local optima and try to find the best possible solution (Bashar & Aldeeb, 2019). The performance of this technique depends upon the stopping criteria and size of the tabu list.

Dewi et al. (2021) proposed a hyper-heuristic approach with HC and TS to optimise the university examination problem using the Toronto dataset. It performs the largest degree algorithm to create an initial solution by creating hyper-heuristics from low-level largest degree heuristics with swap and move operations. In the next phase, it performs HC with low-level heuristics to explore the new feasible solution, if the value of an objective function is less than the previous. It repeats the procedure up to a threshold value. Now finally TS is performed to improve the solution as it avoids the previous search solution to improve the efficiency of the algorithm. He shows the HC greedy approach that quickly improves the fitness of the initial solution. He also compares all three algorithms with each other and found TS is better at optimising the problem and returning an efficient solution as compared to HC and the largest degree algorithm. Overall, his technique performs an improvement of 18%–58% in the initial solution.

Sun and Wu (2023) proposed a solution for UETP for Chinese high schools with an added difficulty of variable classes for college entrance examination preparation and developed a two-phase TS algorithm with a mathematical model with graph colouring to initialise the model and then optimise by performing a neighbourhood search to improve the computational efficiency. The approach is applied to find a feasible solution in implementing a restricted neighbourhood in the first phase by eliminating hard constraints only and then improving the quality of the structure by implementing one-flip and swap operators in the second phase. The algorithm is tested on 15 different benchmark datasets and performs better than the simple MIP technique, but unfortunately, it is not able to solve all of the soft constraints. It shows an average of more than 90% accuracy within the average 38 s. For soft constraint, it shows a more than 71% satisfaction rate using a two-phase Tabu search algorithm.

Simulated annealing

SA was introduced by Kirkpatrick et al. (1983) by developing an analogy of the physical process of heating a metal at a high temperature and cooling off to get a crystalline structure until no more changes occur (Kirkpatrick et al., 1983). It starts with a random solution and iteratively compares it with another random solution by calculating a specific probability between the current and candidate solution to find an optimal solution (Bashar & Aldeeb, 2019).

Leite, Melicio, and Rosa (2019) suggests the FastSA algorithm which is an extension of the SA algorithm to reduce computation time to resolve hard and soft constraints by following saturation degree-based heuristics. If a selected exam has a scheduling conflict, it will be moved. Otherwise, it will remain unchanged. In this way, it will improve the significant performance of the algorithm and show 17% more performance as compared to simple SA with the ITC2007 dataset and 234.7 seconds as compared to 377.2 seconds with simple SA. The main problem with this approach is that it is used to optimise a domain-specific problem and is not able to solve a generic problem like simple SA.

Qu et al. (2009) suggests a solution for uncapacitated UETP using SA by using a different combination of novel neighbourhood structures with tuning principles using artificial training instances. It shows better results on the Carter dataset as compared to a simple SA algorithm and also identifies new neighbourhood structures to optimise the problem quickly. With Car92 dataset, it shows on average 3.8 conflicts as compared to an average of 3.93 conflicts with a simple SA algorithm.

Carlsson et al. (2023) implemented two-stage decomposition with constraint programming, mixed integer programming, and a simulated annealing algorithm to solve examination timetabling problems. They also performed three different techniques with mixed integer programming and a hybrid version of all of these techniques. They found a meta-heuristic approach simulated annealing performed nearly 20% better than mathematical algorithms in most of the tests.

Great Deluge algorithm

It is a modified SA in which any worst neighbourhood structure is accepted if the quality of the solution is decreased by a threshold level (Bashar & Aldeeb, 2019). Initial solution quality is defined at the beginning and it is updated iteratively by calculating the decay rate. The worst solution is accepted if its quality is equal to or less than the threshold value.

Ashis Kumar Mandal and Kendall (2020) tries to schedule partial exams to allocate timeslots and rooms by implementing partial graph heuristic orderings by using a modified form of the Great Deluge algorithm with benchmark datasets ITC2007 and Toronto dataset. It starts with a random solution for a partially selected exam based on graph heuristic orderings and then tries to improve the partial solution with the Great Deluge algorithm. It repeats the procedure by selecting the next set of exams and tries to improve until all exams are scheduled. It tries to minimise the single objective function to optimise capacitated as well as uncapacitated UETP. It shows good performance in both benchmark datasets ITC2007 and Toronto as compared to other approaches which perform optimisation on all of the exams at the same time. It shows 4.58 and 3.82 conflicts with the CAR-S-91 and CAR-F-92 dataset, respectively, and gives improved results as compared to a simple colour heuristic algorithm.

Mandal and Kahar (2020) proposed a combination of graph heuristic and metaheuristic approaches to implement UETP. In the first phase, an initial solution is generated using different graph colouring heuristics and hybrid heuristics with five different meta-heuristic algorithms such as TS, SA, variable neighbourhood search, great deluge algorithm, and late acceptance HC with Toronto and ITC2007 datasets. He showed that hybrid heuristics are effective in generating a feasible solution as compared to simple heuristics and GDA. Late acceptance HC and SA perform better in optimising the problem as compared to TS and VNS. It shows 35.51 and 4.57 conflicts with yor-f-83 and car-f-92 dataset, respectively, with proposed hybrid graph heuristics.

Variable neighbourhood search

In this technique, a number of neighbourhood structures are defined to avoid local optima by changing the neighbourhood with others based on search status. The performance of the techniques largely depends upon the number of neighbourhoods. It is applied by Al-Betar (2021) to solve UETP and shows improved performance in a local search, but it becomes slow as the problem size increased.

Population-based approaches

Population-based techniques try to optimise the number of initial solutions called population, and generate a new population iteratively through selection, mutation and crossover to generate a more fitted population, and return the best possible and optimised solution at the end (Bashar & Aldeeb, 2019). Population-based techniques are effective in complex optimisation problems and can identify the required search regions in search space, but it needs more processing time to analyse large populations (Fong, Asmuni, et al., 2022). Genetic algorithm (GA), Ant Colony Optimisation (ACO) and Particle Swarm Optimisation techniques are common population-based approaches used to optimise examination scheduling.

Genetic algorithm

It is the most popular population-based technique introduced by Holland (1975). It is derived from nature by performing the natural process of selection in living organisms (Bashar &

Aldeeb, 2019). In GA, a population is simply a set of chromosomes with several genes representing timeslots to conduct exams. The initial population is generated randomly and it performs various operations such as selection, crossover and mutation to generate a new set of chromosomes by evaluating its fitness. So fitted chromosomes become part of the new population and are used to generate new generations. The process is iteratively performed to search in global space until the best possible solution is found (Bashab et al., 2020). Simply put, it will return the most fitted individual from the population. One of the major problems with GA is the heavy computational cost.

The GATS_CS timetabling system was developed in Abayomi-Alli et al. (2020) by implementing a genetic algorithm with a Tabu search to optimise the UETP, and found 96.07% and 99.02% accurate results for semester 1 and semester 2, respectively, in optimising several course allocations for examination without exceeding examination hall capacity.

Ant colony optimisation

ACO was introduced by Dorigo et al. (1996) and, as a way to optimise, used a colony of ants as an analogy in the same way that ants define a path with high density to navigate to collect food, and avoid the low-density paths (Bashar & Aldeeb, 2019). In the same way, an initial solution based on heuristics is developed, then identifies the optimised neighbourhood in a way to get maximum performance, and iteratively improves the path by discarding low-optimised neighbourhoods and replacing them with highly optimised neighbourhood structures in performing a global search and avoiding local optima. This technique can easily learn from the graph paths with the highest and lowest optimised structure, so it can easily adjust the graph with maximum return to get the optimal solution. It outperforms the GA and other metaheuristic approaches in getting a high-quality solution with low computational cost (Dorigo & Caro, 1999).

Khair et al. (2019) proposed a population-based ACO to optimise the UETP with the Universiti Sultan Zainal Abidin in Malaysia. A random initialisation is performed to create an initial solution, then it implements the largest degree and largest enrolment heuristics to improve the speed of the algorithm. It also calculates the distance matrix to satisfy soft constraints related to rooms if exams are in multiple rooms or if the exam size is larger than the room capacity. The overall results of this algorithm are somewhat satisfactory but not the best when compared to others.

Harmony search (HS)

This is a metaheuristic algorithm developed by Geem in 2001, based on the process of finding perfect harmony by musicians. It performs searches to find the best solution on the same concept (Dubey et al., 2021). There is no need to define the initial settings of decision variables and only a few variables are required for fine-tuning the optimisation process. So, it is an efficient algorithm that is initiated with simple implementation and optimised in a shorter processing time. It is a preferable choice in various clustering, scheduling and optimisation problems.

Alamri et al. (2020) proposed an optimised HS algorithm based on a simple population-based HS algorithm by performing distributed selections with GA to solve the UETP. After initialising the initial solution, it implements a three-dimensional matrix containing information about the day, timeslot and room to generate a deterministic number to eliminate conflicts and to enhance selection to optimise the solution. All of the constraints are satisfied iteratively by performing genetic operations to check the fitness of the individual members of the population to get the fitted solution. It shows 100% results in optimising the examination timetable for Jadara University as compared to a manual system.

Artificial immune system

It is a process of improving the neighbourhood structures by changing the parameters through learning and extracting the best features to improve it from the previous. It works like an immune system of living organisms where antibodies, antigens and immune cells create more protected and strong organisms to avoid foreign agents (Bashar & Aldeeb, 2019). In the same way, different computational structures are injected into several timeslots to accommodate various constraints to set the exams for different students in such a way that it performs better than the previous one. It iteratively generates new structures by performing selection, cloning or mutation, and evaluates the fitness of the solution after injecting it. It is overall an efficient technique to return the best possible solution in less time and computation as compared to GA (Bashar & Aldeeb, 2019).

Hyper-heuristic approaches

These are ML approaches that automate the process of selecting and combining low-level heuristics and generate new high-level heuristics. This technique is independent of the problem domain, so can be used to solve any type of optimisation problem without tuning the solution space (Bashar & Aldeeb, 2019). It selects low-level heuristics to create high-level heuristics to perform a search at each stage. Constructive hyper-heuristics are generated from the low-level construction heuristics by using fuzzy logic or other ML and AI techniques to develop high-quality hyper-heuristics that perform efficient optimisation of the UETP (Pillay & Banzhaf, 2009). It is also an effective technique to develop a flexible and adaptive university examination timetabling solution.

An approach to generate hyper-heuristics for UETP is implemented in arithmetic heuristics and hierarchical heuristics (Pillay & Ozcan, 2017). For this purpose, genetic programming is implemented to construct arithmetic heuristics, and then these random heuristic combinations can be evaluated to learn about hierarchical heuristics using GA and genetic programming (Burke et al., 2009). The algorithm is tested by a random generation of hierarchical heuristics as well as using GA and genetic programming. These hyper-heuristics are used to solve examination problems as well as a UCTP and return better results than graph colouring heuristics when applied to Toronto and ITC2007 datasets (Bashar & Aldeeb, 2019). It is also found that hierarchical heuristics created by GA or genetic programming perform well compared to random generation.

As already discussed, Dewi et al. (2021) constructed a hyper-heuristic framework using the largest degree of heuristics and implemented it using HC and TS. Authors found that the TS algorithm showed 18%–58% improvement in the initial solution generated with the Toronto dataset.

Use of hybridisation and AI-based techniques in solving UETP via SLR (RQ4)

This section describes one of the other major areas in the research field of UETP to get information about the use of hybridisation and AI in solving the UETP. The use of hybridisation is to combine multiple techniques mentioned in the answer of RQ3 to improve the optimisation algorithm for getting a solution of UETP, whereas the impact of AI-based techniques shows an improved, efficient and robust algorithm for finding the solution.

Hybrid approaches

Hybridisation is a way to design an algorithm to optimise the problem by combining two or more optimisation techniques to overcome the weaknesses of those techniques and to get a quick and reliable solution (Bashab et al., 2020).

Figure 3 describes a hybrid algorithm where a global search is performed by some population-based algorithm, then local searches are performed on the various regions to identify the optimised solution. All of the optimisation techniques are either local search or population-based approaches, have some inherited defects and are not able to get optimised results, so combining these will create an improved and efficient solution for UETP. Hybridisation can be achieved in different ways such as combining different meta-heuristic approaches, integrating domain-specific heuristics, or use of AI and ML to improve the performance of the algorithm and the quality of the initial solution (Bashar & Aldeeb, 2019). Several researchers have proved the effectiveness of examination scheduling by implementing it with a hybrid approach compared to a simple optimisation technique.

Sharma and Sharma (2020) proposes a metaheuristic technique, an intelligent constraint-handling evolutionary algorithm, by optimising partial solutions to satisfy all constraints in identifying a feasible solution for the problem. A generic fitness function is calculated incrementally with constraint programming for different constraints and then refines the solution with an evolutionary algorithm by performing intermarriage crossover with variable-sized chromosomes to adjust dynamic constraints. Constraint programming is implemented to find the optimal structure by minimising the fitness function with crossover and mutation in exploring with a revertible clonal hill-climbing local search. It shows good results with Toronto benchmark datasets to minimise single objective functions to find the solution for UETP.

Sun and Wu (2023) implements GA with linear scalarising and compromise programming to solve UETP to minimise multiple objective functions. Compromise programming is used to set multiple objective functions based on preference and non-preference and perform some fuzzy rules to set balance among multiple objectives. To get overall fitness, a single objective is also defined by using all of the objective functions. A hybrid approach is implemented by initialising the population with a greedy algorithm to generate a feasible solution. Then it iteratively performs different genetic operations with the repair option to find the optimal solution. Research gives a good approach to developing solutions for the preferences of different stakeholders, but it is not efficient in developing a quality solution.

Abayomi-Alli et al. (2020) proposes a hybrid approach that combines a modified GA with TS memory techniques and incorporates course sandwiching to solve the UETP. In the selection process, a random solution is generated by evaluating the fitness with a single objective function to satisfy hard constraints. The optimisation phases are divided into

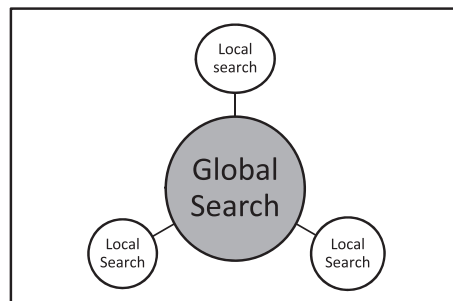


FIGURE 3 Embedding a global search and local search in a simple hybrid algorithm.

Taburisation and Optimisation. In Taburisation, a TS with long and short memory is performed to improve the fitness of the solution, whereas Optimisation adjusts the fitness of the solution by performing a crossover operation. In the fine-tuning phase, a sandwiching operation is performed to merge multiple exams in the same room from the Tabu long memory from the previous phase. In the next section, it separates the exam into allocated days for a period of exam and shifts, and separating the exams that need to be sandwiched is used and evaluation is implemented with GA and TS, and course sandwiching replaced Crossover and Mutation. He shows that the proposed algorithm resolved nearly 96.07% and 99.02% of conflicts accurately for semester 1 and semester 2 respectively from the dataset of the public University in Nigeria, but was unable to resolve all of the constraints. It identifies a hybrid approach provides better results as compared to an individual local search algorithm and genetic algorithm.

Memetic algorithm

This is a population-based hybrid genetic algorithm combined with a local search to satisfy all of the hard and soft constraints. It performs a global search through a genetic algorithm that is fine-tuned by local searches (Ghafar et al., 2021). It is an effective approach to minimise the time for optimising the timetabling solution, but its performance depends upon the local search algorithm used (Burke et al., 1996).

Leite, Fernandes, and Melicio (2018) implements a cellular memetic algorithm to optimise the examination scheduling and shows the impact of the level of threshold metaheuristics in a local search to improve the solution. The effectiveness of his study is also a guide to the balance between exploitation and exploration in implementing hybridisation. A random initial feasible population is created using graph colouring heuristics Saturation Degree. In the next phase, the crossover operator is applied to generate fitted offspring and then the mutation operator is applied with the Kempe chain neighbourhood to improve the fitness of the offspring. Then a local search operator is performed on the fitted offspring to improve its quality by setting a threshold level. It repeats the process for a certain number of times till convergence is achieved. The algorithm is tested with different instances of Toronto and ITC2007 and performs better in many instances compared to traditional evolutionary algorithms.

Li et al. (2018) proposed a memetic algorithm to minimise multiple objectives for UETP based on a multi-objective evolutionary algorithm with decomposition MOEA/D. Hybrid initialisation is performed to create an initial feasible population by using graph colouring heuristics and hyper-heuristics. During initialisation, equally distributed weight vectors are created by showing the distance of each member using Euclidean distance to maintain the distribution of the exam in the overall period. Then the fitness of two random neighbourhoods is iteratively tested. Crossover and mutation operators are performed for unfitted regions. Two different local searches are performed to distribute the exams properly and to improve the quality of the solution. The algorithm does not perform satisfactory results with Carter datasets but it gives a shorter length timetable with a longer period and improves the robust effectiveness of various operators. It shows a 14.8 length period with Era83 dataset as compare to a 11.0 length period with car91 dataset.

Swarm optimisation algorithms

This was developed by Kennedy and Eberhart in 1995 with the concept of particles placed in the search space and evaluating the current location to optimise the best location to move it to another optimal location (Poli et al., 2007). It is a population-based algorithm and defines the global search space that is optimised with local multiple searches. It is an efficient algorithm used in various optimisation and scheduling problems, the same as with timetabling.

Abayomi-Alli et al. (2019) implements a population-based stochastic algorithm, Particle Swarm Optimisation, with a local search to solve the examination problem for the Federal University of Agriculture, Abeokuta, Nigeria. The initial population is created randomly as particles, and these particles are evaluated by their positions and velocities to move toward the fitted solution in search space and update the position and velocities of the particles. After that a local search is applied to allocate the best-fitted rooms or halls for the exams. He shows that PSO with a local search performs with 70%–80% accuracy and better than simple PSO. It shows 84.1% and 76.2% accuracy with semester 1 and semester 2 datasets, respectively, with a local dataset.

Aldeeb et al. (2021) proposed a population-based Intelligent Water Drops algorithm to improve the fitness of UETP. To improve the exploitation process, it performs hybridisation with the local search algorithm. To construct a solution, graph colouring heuristics Saturation Degree is performed after assigning a random slot to a random exam. In hybrid mode, a local search is applied to the constructed solution to improve its quality by implementing it in different neighbourhood structures. In the next phase, a single objective function is evaluated to improve the solution iteratively. The proposed solution performs well for some instances of Carter datasets with other, different optimisation techniques. It also observes that the hybrid IWD performs well compared to the simple IWD algorithm. It shows 5.41 conflicts with Hybrid IWD as compare to 7.82 conflicts using a cars911 dataset with simple IWD algorithm.

Tilahun (2019) proposed the Prey–Predator algorithm, a metaheuristic approach, to optimise the UETP. It creates a random initial solution using graph colouring heuristics and saturation degree with a minimum penalty and is categorized into the best solution as prey, the worst as predator, and others as ordinary prey. It performs a balanced approach of hybridisation by exploiting the prey using neighbourhood structures, while exploring the predator to identify the best solution. Ordinary prey is replaced by the solutions found by exploitation and exploration after minimising the single objective function. Balancing exploration and exploitation are important in achieving the efficiency and performance of the algorithm. The author implements the algorithm with the Carter Uncapacitated dataset and shows the best performance for a few of the instances. It shows 5.91 conflicts with car92 dataset that is less than the 6.2 shown by Carter M.W.

Fong, Leong, et al. (2022) proposed the Black Hole Algorithm, a hybrid solution, to optimise the UETP. It tries to improve the exploitation in the selected regions by avoiding local optima to increase the speed of convergence. It starts with a random initialisation using graph colour heuristics and performs optimising the selected region as a black hole, and compares it with other regions and absorbs other candidate regions as stars. If the fitness value of any star is more than the black hole then it will replace the other. The improvement phase is performed iteratively using haploid crossover to move the star toward the black hole and HC is used to perform a local search by using five different neighbourhood structures with a greedy approach, with a threshold level to improve it. They performed a hybrid algorithm on Carter's uncapacitated dataset to minimise single objective function and found improved results with a hybrid solution. It shows 3.81 and 9.56 conflicts with car92 and hec92 benchmark datasets, respectively, and stands at second position in different hybrid approaches and third position in various meta-heuristic approaches.

Artificial bee colony

This simulates the behaviour of honeybees in optimising a problem and is defined by Karaboga (2005). Bees communicate with each other by waggle dancing to identify the best path as compared to previous paths to find food and to guide others to follow it (Bashar & Aldeeb, 2019). To optimise the timetabling problem, some neighbourhood structures are

randomly defined and then iteratively optimised by exchanging the exams in different slots and comparing the performance with each other to match the best possible solution. It also explores the new optimised structures to avoid local optima and can return the best possible solution. It can find good solutions from a large search space and performs better than other metaheuristic techniques. The Firefly algorithm is a Swarm-based intelligence optimisation algorithm and works on the flashing behaviour of fireflies (Nand et al., 2022). These fireflies use flashing patterns to attract others to help them navigate. It starts with the initial population, then iteratively learns about the brighter fireflies and performs random movement, and ends with the brightest fireflies as the optimal solution. It is a robust optimisation technique and achieves convergence with optimal solutions in less time.

A discrete firefly (DF) algorithm based on intelligent Swarm optimisation is proposed in Nand et al. (2022) to solve the uncapacitated UETP by implementing three neighbourhood searches through heuristic ordering in multiple phases. It performs fitness of the partial exam with heuristic orderings in the discrete domain to optimise the solution step by step. In the first phase, the initial solution is generated by performing the first neighbourhood search using the Largest Degree heuristic. Then two other neighbourhood searches are applied using Kempe Chain heuristics in the next phases to improve the solution by minimising the single objective function. The algorithm performs better not only in exploitation but also in exploration to find the best possible solution. In this way, it can also avoid local optima, and a neighbourhood search returns a more fitted solution. It also performs better due to performing the exams in steps instead of optimising it as a whole. It shows better results with Toronto datasets as compared to the continuous firefly algorithm. It shows 10.17 conflicts with hec92 dataset that is much closer to the lowest previous results.

Ghaisani and Suyanto (2019) proposed a DF algorithm to optimise thesis examination timetabling by discretisation of predetermined constraints to reduce the conflicts, which helps to get quick optimisation. An initial candidate solution is generated randomly using room, time and date. Then distance is calculated to check the fitness of the solution iteratively by adding other attributes to find the fitted solution. It shows 99% accuracy with the proposed algorithm as compared to 70% accuracy with the conventional algorithm on the dataset of Telkom University.

A Swarm-based approach is used to implement the Fruit-Fly Optimisation algorithm to optimise the UETP (Abdul-Rahman et al., 2019). It works on the behaviour of fruit flies to develop a path in search of food. A random solution is created using the largest degree graph colouring heuristics followed by multiple solutions, and optimised by performing four different random neighbourhood structures such as simple swap median exam (SSME), single swap random exam (SSRE), single median random exam (SMRE) and single median median exam (SMME). Then it calculates the fitness of all solutions with the initial solution by using the distance formula, and the most fitted solution is chosen for the next iteration. The performance of the approach largely depends upon the quality of the neighbourhood structure. He performs the algorithm on the dataset of the Universiti Utara Malaysia and identifies the effect of different neighbourhood structures on the final solution. It shows 0.031 and 0.96 standard deviation with SMRE and SSRE, respectively.

Use of AI in a timetabling problem

This section introduces the results of the review to answer the third research question given in this review. AI and ML techniques are the best approaches that can be used to optimise and create an effective solution for any complex problem (Crompton & Burke, 2023). AI techniques are also effective in optimising university examinations timetabling problems and developing a more robust and faster algorithm to get solutions for them. Different AI

techniques such as ML, RL and DL techniques are applied by various researchers to improve the optimisation process in various phases such as initial population generation, local search to avoid local optima, improving the processing of population-based algorithms, as well as developing hyper-heuristics and improving the performance of the overall algorithm (Bashab et al., 2020). ML as well as DL are common AI techniques to develop algorithms to find more efficient and adaptive solutions (Crompton & Burke, 2023). It is also effective to learn from the historical data and to perform constraints accordingly. So, implementing AI in developing a solution for examination timetabling can improve the quality and performance of the solution.

ML and DL approaches can be used to extract the best possible and feasible solution from past data and it will be a good way to generate an initial solution that can be used to optimise using various other meta-heuristic or hyper-heuristic approaches. Use of genetic algorithms and various other evolutionary algorithms to optimise the solution are also based on pure AI and genetic programming (Pillay & Ozcan, 2017). Fuzzy rules are the most important constraints in various optimisation algorithms to define conflicting constraints in a very efficient way and it can also be a good choice to generate an initial solution with little effort (Sun & Wu, 2023). ML, DL, RL and deep RL are sub-domains of AI as shown in Figure 4. These AI techniques have entirely changed the rules of the game and have provided new opportunities in the field of UETP too. Now new research can be performed to create efficient and more optimised algorithm for the problem.

Machine learning

ML is a major AI technique that can be used to learn from an available dataset using various mathematical and statistical models. It can be effectively applied to generate a feasible initial solution for any optimisation algorithm (Sahu et al., 2023). Numerous ML algorithms such as random forest, decision tree, neural network, etc. can be used for this purpose. Artificial Neural network can be used to generate an initial feasible solution from the historical data after about the most optimised features. These techniques depend upon the quality of extracted features,

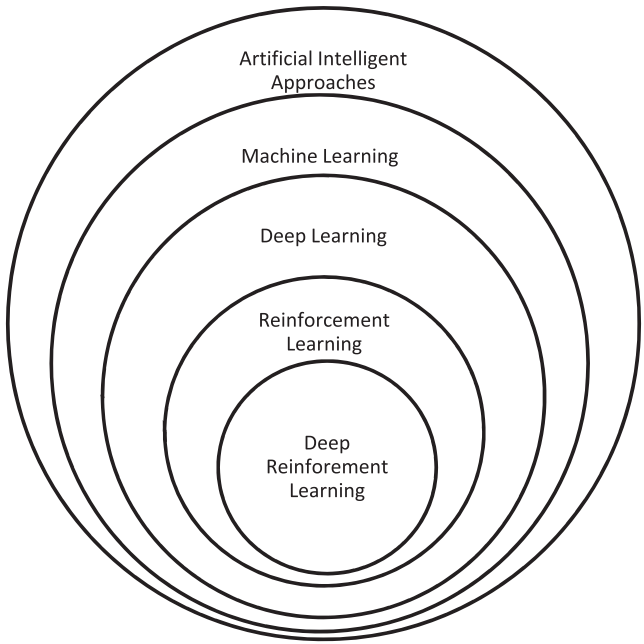


FIGURE 4 Hierarchy of artificial intelligent approaches (Sahu et al., 2023).

so data preprocessing, data cleaning and feature engineering are the most important tasks before application of ML models. ML is already successfully applied in other fields, especially in medical, but now its use in optimisation has shown a positive impact on the performance. Most of the hyper-heuristic approaches perform ML to learn low-level heuristics to create high level heuristics in order to perform robust optimisation for the UETP (Pillay & Banzhaf, 2009).

Deep learning

DL is the advance form of ML, where there is no particular need of feature engineering and feature extraction. It automatically learns the features to develop a model to provide a desirable solution, but it needs large size data to learn the maximum about the data. CNN, RNN and deep neural network – an extended form of artificial neural network – are the most popular deep-learning algorithms (Sahu et al., 2023). It can be an effective approach to learn the successful scheduling and optimisation used in the past and analyse the past historical data, and then develop the rules to schedule and optimise new data for an effective and efficient solution. It is also preferable to work easily with large-size data. So, the future of optimisation algorithms for UETP greatly depends upon the application of deep-learning models.

Reinforcement and deep reinforcement learning

Reinforcement learning (RL) is a ML technique that can be an effective solution for an optimisation algorithm. In this technique, any intermediate solution will select if it is a feasible solution, otherwise it is rejected and tries to find other one. So, it will learn from the feedback of each solution until a solution with the maximum performance is achieved. A ML model is iterated to find the optimal solution. This technique will be an efficient approach in the UETP by generating a timetable and comparing its overall effectiveness with the previous timetable. A new timetable will be selected if its performance is more than the previous timetable. It can be easily implemented to minimise a multi-objective timetabling problem. Deep RL is the more intelligent technique to optimise the problem automatically by performing deep-learning to learn the best features from the historical data and provide the most optimised solution (Sahu et al., 2023).

Pillay and Ozcan (2017) developed two hyper-heuristics by using artificially intelligent approaches to generate an initial solution and then optimising the solution with a genetic algorithm implemented with artificially intelligent techniques to solve the examination timetabling problem with the Toronto and ITC 2007 datasets. He showed satisfactory results of various implemented algorithms that outperformed the existing hyper-heuristic approaches. It shows 28% improvement compared to a simple hyper-heuristic approach using the Toronto dataset, as well as outperforming other heuristic approaches.

Various intelligent techniques are implemented by Aldeeb et al. (2021), Dewi et al. (2021), Fong, Leong, et al. (2022) to develop more intelligent and robust algorithms by applying these intelligent techniques in generating initial search space, optimising the global parameters of population-based algorithms, and improving the search performed by local search algorithms in optimising the timetabling problem.

Pros and cons of these techniques

Based on the discussion in previous sections, all of the algorithms for optimising UETP as discussed in this SLR, are summarised below in Table 9 in terms of their strengths and weaknesses. Graph Colour theory, Mix Integer Programming and Constructive heuristic algorithms are easy to implement and customise with small and medium-size data, but these algorithms become complex and difficult to implement with large size data. Mixed integer programming and constructive heuristic algorithms are based on mathematical equations so they are efficient in developing a mathematical model for the UETP and show a low

improvement rate, even when they have a longer execution time. These algorithms are also suitable for hard constraints but unable to provide an optimal solution for soft constraints. Local search algorithms such as Simulated Annealing, Tabu Search, Hill-climbing, Great Deluge and variable neighbourhood searches provide an optimised local solution but their performance depends upon the initial solution as well as parameter tuning. The Hill-climbing algorithm gets stuck in local optima while finding the optimised solution. Few of these algorithms avoid local optima but there may be a chance of premature convergence in that case. Most of these local search algorithms are slow in convergence due to performing a local search on large-size data. Population-based algorithms such as the Genetic algorithm, Ant Colony Optimisation and HS are suitable for large-size data and performs global search to avoid local optima, but they need high computation resources and are slow in convergence with complex constraints. They are also sensitive to the initial solution as well as tuning parameters setting. Hybrid algorithms such as Swarm optimisation, Artificial Bee Colony and Memetic optimise the global solution with a number of local searches to avoid local optima, but they need heavy computation cost. These are flexible with complex constraints but there are chances of premature convergence due to dependence on the global solution. Optimisation algorithms having AI capabilities such as Genetic algorithm and Ant Colony Optimisation are more efficient even with complex constraint to find an optimised solution. AI techniques can handle a large number of constraints but implementing conflicting constraints is still a big challenge. Artificial intelligent techniques are preferable to create an initial solution and are able to perform automatic error handling in performing global and local searches, but AI algorithms need a more computational cost. ML and DL provide an opportunity to explore historical scheduling data to create new optimised solutions for new situation by minimising single or multiple objective functions. Sometimes these algorithms overfit the historical data to generate a new solution. RL and deep RL are more intelligent techniques to optimise the timetabling problem with maximum performance by handling the number of constraints easily, but these techniques require a lot of processing capabilities and are only preferable for large size data.

Strengths and weaknesses of optimization techniques to solve the UETP are summarised in Table 10. Heuristic approaches are easy and simple to implement with small-size data, but its performance depends upon the initial solution and tuning parameters. Meta-heuristics are able to provide a global solution to avoid local optima, but they also highly depend upon parameter tuning and require a high computational cost. Population-based techniques such as Genetic algorithm, Swarm Optimisation, etc., perform a global search and tune the solution with local searches, but there is more chance of premature convergence. In most of the local search algorithms such as TS, HC, SA, etc., local optima occur. Hyper-heuristic techniques are flexible in heuristic selection and tuning parameters, but it becomes complex with a large number of heuristics and are sensitive to an initial heuristic selection. These techniques can improve and optimise previous solutions to create new solutions, but there is a great chance of overfitting with a poor set of heuristics. Hybrid techniques such as s provide efficient solution by handling complex constraints. It can adjust the global search with local searches to improve the performance of the solution, but the the hybridisation level greatly impacts the performance. Due to the complexity of hybridisation, it becomes difficult to tune the parameters and becomes slow in convergence.

SUMMARY AND DISCUSSION

The review of UETP provides a clear understanding of the problem domain and several constraints associated with the UETP. Table 5 describes the answer to the first research

question and gives an overview of various constraints used in implementing the UETP. Sometimes, some of the software constraints conflict with other constraints and impact the performance of the solution. These conflicting constraints cause a lot of problems in generating smooth timetables, so it is important to understand these constraints to avoid these conflicts. Artificial techniques such as fuzzy logic rules can be used to resolve and implement these conflicts easily but it may impact computation cost. DL and RL can also provide a better solution for software constraints by analysing the historical and comparing the previous solution. Conflicting software constraint and a large number of constraints slow down the processing of AI techniques as well. We can see in [Table 5](#), these constraints can be categorised based on the stakeholders involved in getting the solution for UETP such as students, teachers, invigilators and management. It will be the answer to the second research question to get the optimised solution for one or more stakeholders to construct an optimised UETP for all of the stakeholders. Single-objective UETP optimisation approaches provide a single comprehensive solution irrespective of any particular stakeholder, whereas the multi-objective UETP optimisation approach provides a global solution that tries to satisfy all of the stakeholders. In [Table 8](#), one can see that multi-objective optimisation algorithms are better than solutions having a single-objective function. More than 75% of research discussed in this SLR is performed using single-objective function due to simplicity and less processing requirements. In the same way, most of the available benchmark datasets for UETP are also inclined to achieve a single-objective rather than optimising multiple objectives (Ceschia et al., [2023](#); Qu et al., [2009](#)). So, there is a huge gap in performing research to optimise multi-objective UETP as well as developing a dataset for it.

Several optimisation techniques such as heuristics, meta-heuristics, hyper-heuristics and AI techniques can be applied to solve the UETP. Due to the complex nature of the problem, all of these techniques give near-to-optimal solutions rather than achieving 100% accuracy. So, designing or developing a new algorithm that shows an improvement in the proposed solution for the UETP, is still a research topic. The answer to the third research question is deeply explored to understand these techniques and algorithms and understand the ways to implement and improve the optimal solution for the UETP. [Table 8](#) also describes the description of work performed in optimising the solution for examination scheduling. [Table 11](#) describes various algorithms to optimise the UETP and also shows the performance of these algorithms and techniques. During this review, one can easily identify research gaps in this field of university examination timetabling and is able to improve the accuracy of the implemented algorithms by following future recommendations.

[Table 8](#) shows that graph colouring is still popular to formalise UETP before implementing any optimisation algorithm. The resultant graph is easy to convert into a conflict matrix to be used to resolve conflicts in optimisation algorithms, whereas mathematical models or constraint programming is a better choice for MIP optimisation (Keskin et al., [2018](#)). One can see the MIP technique is still the better choice for UETP optimisation due to mathematical equations which are easy to implement and helps to give a quick and robust solution (Ajik et al., [2023](#)), but it requires a lot of processing capabilities and time, even for a small problem.

In previous sections, we also learned that the result of any optimisation approach used to solve UETP largely depends upon the initial solution. An initial solution with a large number of conflicts can never give an efficient solution at the end. For this purpose, most of the optimisation algorithms discussed above, perform repairing of the initial solution (Abdul-Rahman et al., [2019](#); Aldeeb et al., [2021](#); Bellio et al., [2021](#)) or perform some intelligent approach to create a feasible solution (Pillay & Ozcan, [2017](#)). Several techniques are used to generate an initial optimised population for population-based techniques (Abayomi-Alli et al., [2020](#)). One of the favourite approaches in the past was graph heuristics, which is still widely used to initialise the initial population for various heuristics, metaheuristics (Leite,

TABLE 8 Algorithms and features of UETP approaches in selected papers.

Algorithm	Type	Objective	Model	Initialisation	Approach type	Approach	Algorithm	Dataset
Cellular memetic algorithm (Leite, Fernandes, & Melicio, 2018)	Capacitated Uncapacitated	Single	Graph colouring	Random	Population-based	Hybrid	Memetic Algorithm	Toronto ITC-2007
Fast simulated annealing with threshold acceptance (Leite, Melicio, & Rosa, 2019)	Capacitated	Single	Graph colouring	Feasible Solution	Single-Solution	Single solution	Simulated Annealing Algorithm	ITC2007
MOEA/D memetic algorithm with Local Search (Li et al., 2018)	Uncapacitated	Multiple	Heuristics	Hybrid Solution	Population-based	Hyper-Heuristics Hybrid	Memetic Algorithm	Toronto
Integer linear programming with multiple phases (Al-Hawari et al., 2020)	Capacitated	Single	Mathematical	Random	Single-Solution	Heuristics	Mixed integer programming	Custom
Mixed integer programming in Excel (Gular & Gecici, 2020)	Capacitated	Multiple	Mathematical	Random	Single-Solution	Heuristics	Mixed integer programming	Custom
Mixed integer programming (Guler et al., 2021)	Capacitated	Single	Mathematical	Random	Single-Solution	Heuristics	Mixed integer programming	Custom
B-hill climbing algorithm (Al-Betar, 2021)	Uncapacitated	Single	Graph colouring	Random	Single-Solution	Single solution	Hill Climbing Algorithm	Toronto
Partial search great deluge algorithm (Ajik et al., 2023)	Capacitated Uncapacitated	Single	Graph colouring	Hybrid Solution	Single-Solution	Single solution	Great Deluge Algorithm	ITC2007 Toronto
Ant Colony Optimisation (Dorigo & Caro, 1999)	Capacitated	Single	Mathematical	Random	Population-based	Meta-heuristics	Ant Colony Algorithm	Custom
Preference-based stepping Firefly algorithm (Nand et al., 2022)	Uncapacitated	Single	Graph colouring	Hybrid Solution	Population-based	Meta-heuristics	Firefly Algorithm	Toronto
Graph colouring (Oke et al., 2023)	Capacitated	Single	Graph colouring	Random	Single-Solution	Heuristics	Graph Colouring Theory	Custom

TABLE 8 (Continued)

Algorithm	Type	Objective	Model	Initialisation	Approach type	Approach	Algorithm	Dataset
Two-stage heuristic integer programming (Keskin et al., 2018)	Capacitated	Single	Mathematical	Random	Single-Solution	Heuristics	Mixed integer programming	Custom
Fruit-fly evolutionary algorithm (Abdul-Rahman et al., 2019)	Capacitated	Single	Graph colouring	Feasible Solution	Population-based	Meta-heuristics	FruitFly Algorithm	Toronto
Genetic algorithm with Tabu Search (Abayomi-Ali et al., 2020)	Capacitated	Single	Mathematical	Random	Single-Solution	Hybrid	Genetic Algorithm	Custom
Genetic algorithm with fuzzy rules (Tung et al., 2021)	Capacitated	Multiple	Mathematical	Feasible Solution	Population-based	Meta-heuristics	Genetic Algorithm	Custom
Hybrid – intelligent water drops (Aldeeb et al., 2021)	Uncapacitated	Single	Mathematical	Random	Single-Solution	Hybrid	Intelligent Water Drop	Toronto
Mix integer programming (Mustafa et al., 2022)	Capacitated	Single	Mathematical	Random	Single-Solution	Heuristics	Mixed integer programming	Custom
Mix integer programming to Optimising Two-in-row and Two-in-day lower bounds soft constraints (Arbaoui et al., 2019)	Capacitated	Single	Mathematical	Random	Single-Solution	Heuristics	Mixed integer programming	ITC2007
Integer programming (Godwin, 2022)	Capacitated	Multiple	Mathematical	Feasible Solution	Single-Solution	Heuristics	Mixed integer programming	Custom
Discrete prey–predator algorithm (Tliahun, 2019)	Uncapacitated	Single	Graph colouring	Random	Population-based	Meta-heuristics	Pre-predator Algorithm	Toronto
Mix integer programming (Rosyidi et al., 2019)	Uncapacitated	Multiple	Mathematical	Feasible Solution	Single-Solution	Heuristics	Mixed integer programming	Custom
Tabu search and hill climbing (Dewi et al., 2021)	Capacitated	Single	Mathematical	Random	Single-Solution	Hyper-Heuristics	Tabu Search/Hill Climbing	Toronto

(Continues)

TABLE 8 (Continued)

Algorithm	Type	Objective	Model	Initialisation	Approach type	Approach	Algorithm	Dataset
Two-Stage Multi-neighbourhood Simulated Annealing Algorithm (Bellio et al., 2021)	Uncapacitated	Single	Mathematical	Feasible Solution	Single-Solution	Meta-heuristics	Simulated Annealing Algorithm	Toronto
Hill Climbing Hybrid black hole algorithm (Fong, Leong, et al., 2022)	Uncapacitated	Single	Graph colouring	Random	Single-Solution	Hybrid	Black Hole Algorithm	Toronto
Mix integer programming (Siddiqui et al., 2018)	Capacitated	Multiple	Mathematical	Random	Single-Solution	Heuristics	Mixed integer programming	Custom
Optimised Harmony Search Algorithm (Alamri et al., 2020)	Capacitated	Single	Heuristics	Hybrid Solution	Population-based	Hybrid	Harmony Search Algorithm	Custom
Tabu Search/Simulated Annealing/Great Deluge/Late Acceptance HC/Variable Neighbourhood Search (Mandal & Kahar, 2020)	Capacitated Uncapacitated	Single	Heuristics	Hybrid Solution	Single-Solution	Meta-heuristics	Tabu Search/ Simulated Annealing/ Great Deluge/Late Acceptance HC/Variable Neighbourhood Search	Toronto ITC2007
PSO Local Search Algorithm (Abayomi-Alli et al., 2019)	Capacitated	Single	Mathematical	Random	Single-Solution	Hybrid	Particle Swarm Algorithm	Custom
Hierarchical Heuristic Genetic Algorithm (Pillay & Ozcan, 2017)	Capacitated Uncapacitated	Single	Heuristics	Hybrid Solution	Single-Solution	Hyper-heuristics AI	Genetic Algorithm/ Genetic Programming	Toronto ITC2007
Intelligent constraint handling evolutionary algorithm (Sharma & Sharma, 2020)	Capacitated	Single	Mathematical	Hybrid Solution	Single-Solution	Heuristics	Mixed integer programming	Toronto

TABLE 8 (Continued)

Algorithm	Type	Objective	Model	Initialisation	Approach type	Approach	Algorithm	Dataset
Discrete Firefly Algorithm (Ghaisani & Suyanto, 2019)	Capacitated	Single	Mathematical	Random	Population-based	Meta-heuristics	Firefly Algorithm	Custom
Two-phase Tabu Search (Sun & Wu, 2023)	Capacitated	Single	Graph colouring	Random	Single-Solution	Meta-heuristics	Tabu Search	Custom
Two-Stage Decomposition (Carlsson et al., 2023)	Capacitated	Single	Mathematical	Random	Single-Solution	Meta-heuristics	Simulated Annealing	Custom
Profiling-based Algorithm (Alrawasdeh et al., 2023)	Capacitated	Single	Mathematical	Feasible Solution	Single-Solution	Heuristics	Constraint Programming	Custom
Scheduling with Graph Colouring Theory (Oke et al., 2023)	Capacitated	Single	Graph Colouring	Feasible Solution	Single-Solution	Heuristics	Graph Colouring Theory	Custom
Integer Programming with Time Window (Kasm et al., 2019)	Capacitated	Single	Graph Colouring	Feasible Solution	Single-Solution	Heuristics	Integer Programming	Custom
Heuristic Algorithm using Sudoku grid (Alkallak et al., 2022)	Capacitated	Single	Mathematical	Feasible Solution	Single-Solution	Heuristics	Constructive Heuristic Approach	Custom

Melicio, & Rosa, 2019) and hybrid algorithms (Nand et al., 2022). It can also be merged with artificial intelligence techniques in different ways to create a hybrid initialisation (Li et al., 2018) and the construction of hyper-heuristics that gives a more quality solution for UETP (Pillay & Ozcan, 2017) as shown in Table 8.

Table 9 explains the pros and cons of all the algorithms discussed in this SLR. All algorithms implementing graph colouring technique or mixed integer programming are easy to implement but unable to work with large-size data. All population-based algorithms can perform global searches to avoid local optima, but become complex with large-size data and more computational resources required for it, whereas local search algorithms normally get stuck in local optima but are able to generate a more optimal solution in a limited domain. The hybrid approaches are able to manage complex constraints as well as avoid local optima easily, but to set the level hybridisation is difficult. They also take a long processing time in handling large-size data. A hyper-heuristic approach can provide solutions for complex problems with a number of diversified hard and soft constraints easily, but they require an extra computational cost. Artificial intelligent techniques are able to handle large-size and complex data and are able to produce solutions using historical past data by using ML, but care is required to avoid overfitting of the solution. These techniques can generate a more feasible initial solution using ML that can improve the performance of any type of optimisation algorithm.

To optimise the UETP, different meta-heuristic approaches as well as hybrid approaches are more famous techniques, but still heuristic approaches are popular due to their mathematical background and efficiency in processing as shown in Table 6. One of the major problems with heuristic approaches is inefficiency in generating an optimal solution as compared to meta-heuristic and hybrid approaches, as these are slow in processing and takes more time to generate a solution as shown in Table 10. Population-based meta-heuristic approaches are not effective in giving feasible solutions without performing a local search (Abayomi-Alli et al., 2020; Li et al., 2018). On the other hand, the algorithms that perform only local searches are also not efficient in returning the best solution (Al-Betar, 2021; Sun & Wu, 2023). Table 10 shows that hybrids and algorithms with AI capabilities perform better than heuristic and meta-heuristic approaches. One of the major defects of heuristics and meta-heuristics is the inability to deal with large size data and to recover from a local optima situation. In conclusion, all of the approaches have their pros and cons and they are not able to provide the best possible optimal solution. Every approach is sensitive to the initial feasible solution as well as parameters that are required to initialise before its execution.

One can understand the importance of local search techniques combined with population-based approaches in getting a quality solution for the UETP (Aldeeb et al., 2021; Tilahun, 2019). All of the population-based strategies such as evolutionary algorithms never return better results if implemented without local searches (Nand et al., 2022). Local search algorithms may skip a lot of the best implementations of the same problem. To achieve the benefits of population-based as well as local search algorithms, a hybrid solution is an important choice to initiate any new development or extension of previously available algorithms. The research also shows that a hybrid approach will always be a good choice to optimise UETP with a more efficient and feasible solution (Leite, Fernandes, & Melicio, 2018; Sun & Wu, 2023).

Hyper-heuristics and AI techniques were not applied to a great extent in the past due to their complexity and difficulty to implement, but these could perform better results as compared to other meta-heuristics and simple hybrid solutions as shown in Table 6 (Pillay & Ozcan, 2017). ML can be applied to set the level of hybridisation by setting the right mix of population-based and local search techniques. The answer to the fourth research question can be observed in Table 11 if these approaches are embedded with AI techniques, when they will perform better than simple approaches (Aldeeb et al., 2021), as well as every new

TABLE 9 Pros and cons of various optimisation algorithms implemented to solve UETP.

Algorithms	Strengths	Weaknesses
Graph Colouring Theory (Bashar & Aldeeb, 2019; Tilahun, 2019)	- It is simple and easy to implement - Easy mapping of hard constraint	- It becomes complex for large number of constraints - Difficult to implement soft constraints
Mixed integer programming (Keskin et al., 2018)	- Easy to customise to get exact solution - It is flexible to adjust the required solution - It is based on mathematical equations.	- It becomes complex for a large size problem - Low improvement rate with increase in execution time
Constructive Heuristic Algorithm achieved (Li et al., 2018; Pillay & Banzhaf, 2009)	- It is simple to implement - It provides a quick solution then improves it incrementally - It provides a feasible solution satisfying hard constraints	- Depends upon initial solution - It provides a local optimal solution - Unable to satisfy all soft constraints
Simulated Annealing Algorithm (Kirkpatrick et al., 1983; Leite, Melicio, & Rosa, 2019)	- It avoids local optima - Flexible to adjust number of constraints	- It becomes slow due to heavy computation - Its performance depends upon parameter tuning
Hill-Climbing Algorithm (Al-Betar, 2021)	- Quick solution small size problem - Low computational cost	- Its performance depends upon initial solution - Sensitive to threshold value - It is normally stuck in local optima
Great Deluge Algorithm (Bashar & Aldeeb, 2019)	- It is adaptable to customise its parameters - It can avoid local optima easily	- Largely sensitive to threshold value - Depends upon initial solution - It converges slowly
Variable Neighbourhood Search (Al-Betar, 2021)	- It restricts local optima - It is good for large size problem - It handles complex constraints easily	- It depends upon neighbourhood size - It becomes slow for large size problem
Tabu Search (Bashar & Aldeeb, 2019; Dewi et al., 2021)	- It can handle large size dataset easily. - It prevents local optima by avoiding the visited data	- It requires tuning of a number of parameters - Need high computational resources
Genetic Algorithm (Bashab et al., 2020)	- It performs global search to avoid local optima - It is suitable for large size problem - It is flexible to adjust constraints	- it needs high computation - It requires tuning of parameters to a high extent
Harmony Search Algorithm (Dubey et al., 2021)	- It performs global search and avoids local optima - It performs effectively with large size data	- It converges slowly - Performance depends upon local search - It is not efficient with complex constraints
Artificial Bee Colony Algorithm (Dorigo & Caro, 1999)	- It performs a global search and avoids local optima - It can adapt various constraints in getting optimal solution - It is more effective for a multi-objective problem	- It becomes slow with a large size and complex problem - Its performance depends upon parameters setting - Imbalance between global and local searches - Chances of premature convergence

(Continues)

TABLE 9 (Continued)

Algorithms	Strengths	Weaknesses
Swarm Optimisation Algorithm (Poli et al., 2007)	• It performs a global search with multiple local searches to avoid local optima • It requires few parameters settings • Flexible to apply with different constraints	• Its performance is impacted by initial parameter settings • Chance of premature convergence
Memetic Algorithm (Burke et al., 1996)	• Improves performance as able to perform global and local searches • It adapts any local search algorithm to handle different constraints • It identifies regions with global search then performs a local search to optimise it	• Its performance largely depends upon a local search algorithm • Chance of premature convergence • Need more computation for large size problem

technique used to optimise the UETP gives more accurate results than previously available techniques. It also clearly explains that heuristic approaches generate efficient solutions rather than integer programming or graph colouring techniques. In the same way, meta-heuristic approaches perform better than simple heuristic approaches. Hybrid approaches as well as algorithms performing any activity using AI techniques give a more accurate and optimised solution than heuristic or meta-heuristic approaches. Hyper-heuristic approaches outperform the meta-heuristic approaches. Hybrid approaches outperform simple population-based or local search algorithms. Implementing a hyper-heuristic approach using ML outperforms simple hyper-heuristic approaches. Even the performance of the optimisation algorithm can be improved just by adjusting any partial function of the algorithm through using AI-based processing.

Several AI techniques such as fuzzy logic, genetic programming, etc. can perform different operations to improve the optimisation algorithm (Sharma & Sharma, 2020). Fuzzy logic can be used to construct hyper-heuristics as well as perform effective neighbourhood searches to find the best solution with minimum time. Fuzzy logic provides a simple mechanism to create rules for constraints to be implemented to optimise the UETP. So, these AI-based approaches such as genetic algorithms or evolutionary computations improve the performance of the population-based algorithms. Use of AI techniques performs various operations in hybrid algorithms to make it more productive in convergence. These techniques can also be used to set the right mix of hybridisation to adjust the population-based and local search algorithms. Constructive heuristics is another modern and efficient approach to developing high-level heuristics by using ML techniques from low-level heuristics and then used to design an algorithm that returns a more efficient and robust solution (Pillay & Ozcan, 2017). They are also efficient in optimising soft constraints with less time as compared to simple meta-heuristic approaches. Most of the heuristic and meta-heuristic approaches are not efficient with software constraints, but handling software constraints is effective with AI and ML techniques.

The performance and accuracy of various algorithms discussed in this SLR are also observed in Table 11. None of the algorithms ever return a 100% performance except a few of the searches where local data was used to generate a custom solution for the UETP. In reality, it is not possible for any type of algorithm to generate a fully optimised solution but provides a near optimal solution. It can be observed that hybrid approaches and use of AI techniques such as ML will provide more efficient and more optimised solutions than

TABLE 10 Pros and Cons of various optimisation techniques in UETP.

Technique	Strength	Weakness
Heuristic Approaches	• These are easy and simple to implement • These techniques are efficient with small-size data	• They normally get stuck in local optima • Parameter tuning becomes complex to optimise the solution • Performance of the algorithm depends upon the initial solution • Solution is biased toward defined constraints
Meta-Heuristic Techniques	• Provide global solution to avoid local optima • These provide an efficient solution with large-size data • These are flexible and adaptable with complex constraints	• They require a high computation cost • Chances of premature convergence • Parameter tuning becomes complex
Hyper-Heuristic Techniques	• They are more flexible with a selection of heuristics • They automatically tune the parameters • These will improve and optimise the previous solution	• It becomes complex and requires more computation cost • Performance of solution depends upon initial set of heuristics • Chances of over-fitting with poor set of heuristics
Hybrid Algorithms	• It provides a more efficient solution • It can handle complex constraints easily • It optimises global search segments with local searches to avoid local optima	• Performance impacted by level of hybridisation • It needs more computation with large size data • Parameters tuning becomes complex to adjust hybridisation • It is slow in convergence
Artificial Intelligent Techniques	• It can deal with complex and large number of constraints • It can implement historical data to generate a new optimised solution • It can generate more feasible initial solution and handle complex optimisation process	• Unable to deal with conflicting constraints • Chance of overfitting of historical data • It needs more computational cost

meta-heuristic and heuristic approaches. It is observed that the use of AI techniques can improve the time to generate a feasible and optimal solution, able to process a greater number of conflicts as compare to other old techniques, and can perform efficient searches with large-size data.

It is also found in Table 4 that most work in the field of UETP is related to capacitated examination timetabling as compared to uncapacitated UETP. Uncapacitated UETP is not a real problem in the universities but still, it is a hot research topic to improve the performance of optimisation algorithms in the domain of timetabling problems. Several researchers implement both of the scenarios in designing an algorithm to optimise both of them.

Table 4 also shows the use of different datasets in optimising the UETP. Most of the available solutions in the reviewed literature are tested with famous benchmark datasets such as Toronto and ITC 2007. It is also found that most of the research is performed in a lab only, so there is room to implement a solution for commercial use. Most of the researchers try to optimise the timetabling problem using their custom datasets, but its validity is not confirmed. Instead, those are good contributions in developing and enhancing algorithms to optimise UETPs.

TABLE 11 Brief description of implemented algorithms with their performance.

Algorithm	Main focus of study	Strength	Performance
Cellular Memetic algorithm (Leite, Fernandes, & Melicio, 2018)	This study helps to understand the balance of exploitation and exploration in hybridisation	Outperforms traditional evolutionary algorithms	4.31 and 3.68 conflicts with car91 and car92, respectively
Fast simulated annealing algorithm (Leite, Melicio, & Rosa, 2019)	It explores new neighbourhood structures to optimise the problem	Outperforms simple SA	17% improvement, compared to simple SA
Multi-objective memetic algorithm (Li et al., 2018)	It helps to perform hybrid initialisation and to optimise multi-objectives by decomposition	Improve initial solution with hyper-heuristics Return shorter length timetable	14.8 length period with Ear83 dataset
Phase-wise integer linear programming algorithm (Al-Hawari et al., 2020)	It suggests phase-wise integer linear programming to improve the performance	Reduce computation cost	2.2 students having two exams in a day with Toronto-e92 dataset
Mixed integer programming (Gular & Gecici, 2020)	It guides on how to solve a problem in Microsoft Excel	Outperforms simple MIP	0.54 conflicts as compared to 0.70 conflicts in manual timetabling
Decomposition method with MIP (Guler et al., 2021)	It guides on how to use a mathematical equation with a decomposition method to optimise the solution	Outperform Simple MIP	Less than 2 min to solve the problem
B hill-climbing algorithm (Al-Betar, 2021)	It guides on how to perform a feasible initial solution by using graph colouring heuristics	Higher performance than simple HC	157.07 conflicts with STA-F-83-1 dataset
Partial graph heuristics with a great deluge algorithm (Ashis Kumar Mandal & Kendall, 2020)	It helps to use partial graph heuristics to initialise and optimise with a partial timetable	Outperforms a simple graph heuristic algorithm	4.58 and 3.82 conflicts with CAR-S-91 and CAR-F-92 dataset, respectively
Ant Colony Optimisation with neighbourhood structure (Khair et al., 2019)	It guides to implement neighbourhood structures to improve speed	Outperform simple meta-heuristic techniques	1.61 average conflicts with FIC30_A1 datasets
Preference-based stepping Firefly algorithm (Nand et al., 2022)	It guides about partial graph heuristics in initialisation and partial optimisation	Outperforms simple dFA	10.17 conflicts using HEC92 dataset with dFA-Step algorithm
Graph coloring (Rahani et al., 2021)	It implements heuristics in sequence to optimise	Performs better than manual timetabling	8 days to complete all exams of 7 subjects
MIP with two-stage heuristics (Bello et al., 2021)	It guides on how to use a two stage heuristic approach with mixed integer programming	Outperforms simple MIP	3.8 average conflicts with car92 dataset

TABLE 11 (Continued)

Algorithm	Main focus of study	Strength	Performance
Fruit-fly evolutionary algorithm (Abdul-Rahman et al., 2019)	It guides on repairing of initial solution and impact of neighbourhood structures on solution	Improves performance of population-based algorithm	0.031 and 0.96 standard deviation with SMRE and SSRE, respectively
Genetic algorithm with Tabu Search (Abayomi-Ali et al., 2020)	It helps to optimise the solution with GA and Tabu Search with phase-wise approach	Outperforms simple GA and local search	96.07% and 99.02% constraints resolved for semester 1 and semester 2, respectively
Genetic algorithm with multi-objective fuzzy rules (Tung et al., 2021)	It suggests compromise programming with a hybrid approach for a flexible solution	Outperforms simple meta-heuristic techniques	98% students completed exams within 4 days
Hybrid – intelligent water drops (Aldeeb et al., 2021)	It guides to improve performance with a local search in traditional Water Drop algorithm	Outperforms simple IWD	5.41 conflicts with cars911 dataset
Mix integer programming (Mustafa et al., 2022)	It suggests MIP with heuristics to improve efficiency and speed	Performs better than manual timetabling	86% improvement
Mix integer programming (Arbaoui et al., 2019)	It guides about lower bound for soft constraint to improve performance	Performs better than simple MIP	29 and 56 cost with yue2013 and yue 2023, respectively, for single exam in a day
MIP with multi-objective functions (Godwin, 2022)	It helps to implement MIP to achieve multi-objective functions	Outperform simple MIP with single objective	4028, and 0.016 s to solve student and faculty objective function
Hybrid discrete prey–predator algorithm (Tilahun, 2019)	It guides to maintain a balance of hybridisation to get an efficient solution	Outperforms simple meta-heuristic techniques	5.91 conflicts with car92 dataset
MIP with multi-objective functions (Rosyidi et al., 2019)	It helps to use a mathematical equation for multi-objective optimisation	Performs better than manual timetabling	29.6 assignments per lecturer as compared to 16.8 assignments per lecturer
Hybrid with Tabu search and hill-climbing (Dewi et al., 2021)	It constructs hyper-heuristic framework with largest degree heuristic, optimises with two local search algorithms	Outperforms simple Hill-climbing algorithm	18%–58% improvements
Two-Stage Simulated Annealing Algorithm (Bellio et al., 2021)	It is used to learn initialisation repairing and to implement multi-objective with two-stage heuristics with SA algorithm	Performs better than simple SA algorithm	4.2 and 3.6 conflicts with car91 and car92 datasets

(Continues)

TABLE 11 (Continued)

Algorithm	Main focus of study	Strength	Performance
Hill-Climbing with Hybrid black hole algorithm (Fong, Leong, et al., 2022)	It guides to improve exploitation to avoid local optima	Outperforms Simple meta-heuristic techniques	3.81 and 9.56 conflicts with car92 and hec92 datasets, respectively
MIP with multi-objective functions (Siddiqui et al., 2018)	Use of mathematical equations to minimise multi-objective functions	Performs better than manual timetabling	8%–14.6% improvement against manual timetabling
Optimised HS Algorithm (Alamri et al., 2020)	It performs the population-based algorithm to optimise the solution	Performs better than manual timetabling	100% allocation of all exams for local dataset
Hybrid Optimisation Algorithm (Mandal & Kahar, 2020)	It guides to use three-dimensional matrix to optimise solution quickly	Performs better than simple graph colour heuristic	35.51 and 4.57 conflicts with yor-f-83 and car-f-92 dataset, respectively
Hybrid with PSO and Local Search (Abayomi-Alli et al., 2019)	It helps to use hybrid initial solution and hybrid algorithm with local search	Outperforms simple PSO and local search	84.1% and 76.2% accuracy with semester 1 and semester 2 datasets, respectively
Hyper-heuristics with Genetic Algorithm (Pillay & Ozcan, 2017)	It guides about genetic programming to construct hyper heuristics	Outperform simple meta-heuristic techniques	28% improvements
Hyper-heuristics with Genetic Programming (Sharma & Sharma, 2020)	It guides to use a local search with PSO to improve its performance	Outperforms GA with single objective	4.91 and 5.04 conflicts with car91 and car92 datasets, respectively
Discrete Firefly Algorithm (Ghaisani & Suyanto, 2019)	It guides to optimise partial solutions with constraint programming with modified hill-climbing	Outperforms simple HC	98.98% accuracy with proposed algorithm as compared with 70% with simple algorithm
Two-phase Tabu Search (Sun & Wu, 2023)	It performs the local search with two phase approach	Performs better than MIP	90% average accuracy in average 38s
Memetic algorithm (Leite, Fernandes, & Melicio, 2018)	It guides to discretise the predetermined constraints to get a quick solution	Outperforms simple population-base and local-search algorithms	99% accuracy
Fast simulated annealing algorithm (Leite, Melicio, & Rosa, 2019)	It implements MIP with multiple phase Tabu Search to improve quality of solution	Outperforms simple MIP	234.7 s as compare 377.2 with simple SA

TABLE 11 (Continued)

Algorithm	Main focus of study	Strength	Performance
Hybrid Algorithm (Carlsson et al., 2023)	It implements constraint programming, MIP and SA algorithm by decomposing timetabling problem	Outperforms simple heuristic approaches and MIP	20% improvement
Constraint Programming Algorithm (Alrawashden et al., 2023)	It performs constraint programming by grouping together the exams with similarities and then scheduling them in available rooms	Performs better than manual timetabling	0.124% conflict for two exams in same day
Graph Colouring Theory (Oke et al., 2023)	It implements Graph colouring theory to schedule exams in available exam slots	Performs better than manual timetabling	100% accuracy for no two exams in a day with local dataset
Integer Programming (Kasm et al., 2019)	It suggests integer programming to solve conflicts within a time window	Performs better than simple integer programming	0.11 s execution time with Masdar_sem_M4 dataset
Constructive heuristic Algorithm (Alkallak et al., 2022)	It implements an intelligent constructive heuristic approach using Sudoku grid	Outperforms simple heuristic approach	100% accurate for 27 students in a classroom

FUTURE WORK

Benchmark datasets are good for developing efficient algorithms and techniques for lab testing in resolving hard and soft constraints in a proposed algorithm, but in real applications, there are a number of different hard and soft constraints that must be satisfied to get an optimised real timetable problem for any particular institute. One of the major contributions in this field will be developing a generic algorithm with comprehensive dataset that can solve all of the possible hard and soft constraints for the custom requirements of any individual stakeholder. To maintain the efficiency of the algorithm, the selection of desired hard and soft constraints must be performed by integrating and developing a new dataset before the execution of an algorithm by incorporating different AI techniques such as fuzzy logic, neural network, ML, RL or DL.

Numerous searches are performed to optimise university examination timetabling using heuristic, meta-heuristic, hyper-heuristic or hybrid algorithms. These algorithms also return the best available optimised solution but are mostly unable to return 100% accurate results. Therefore, more work is required to develop more sophisticated approaches using these approaches embedded with ML and AI techniques. The use of MIP is still a good choice, due to its mathematical base, in developing hybrid algorithms for optimising examination scheduling. Performance of final solutions largely depends upon an initial feasible solution, so quality of an initial solution is also important in optimising any algorithm. For this purpose, constructive heuristics and different AI techniques can also be used. Use of DL approaches would be more effective in generating an initial solution from the large-size dataset.

We have also learned that any meta-heuristic algorithm returns the best result if performed in phases or tuned to the partial solution. These algorithms may be implemented with small variations to apply in a particular order for improved performance. RL and deep RL are recommended to improve the performance of partial algorithms due to having the ability of improving the solution by using a previous outcome. Implementing these algorithms will reduce the computation cost and improve the execution speed and performance as well. Use of these AI techniques is one of the major areas to explore in the field of UETP in order to create more robust and intelligent algorithms that can generate an optimised timetable to handle any complex situation.

Performing population-based algorithms affects the quality of the solution, whereas extensive use of a local search can affect the performance. So, a hybrid algorithm is highly recommended to balance both of the techniques. The use of AI techniques especially DL in hybrid algorithms will also improve the performance and efficiency of the solution. How much hybridisation is required is also a major open question in the optimisation problems. For this purpose, various ML or DL techniques will be implemented to set the right mix of global search with local searches to improve the solution at maximum level. Intelligent hyper-heuristic approaches can be used to develop hybrid algorithms with embedded ML and DL in solving UETP with the perfect balance between population-based algorithms and local search algorithms. Local search and neighbourhood search techniques can provide an optimised solution due to fuzzy logic, ANN, genetic operators, dynamic initialisation, etc. Global search algorithms can be improved with large-size data by implementing AI techniques in performing intelligent searches.

We have just started using ML, DL, RL and deep RL in the field of the UETP. There is a huge opportunity to implement these techniques to get more optimised and intelligent solutions. Normally, one solution set is not preferable to another dataset, but with the help of ML and DL it will be possible to apply it to another dataset. Use of DL will improve the global search by exploring the historical past data and can easily identify the regions with maximum performance. Then it becomes easy to apply local searches having ML support to identify the optimised solution for the problem. RL has opened new challenges in the field

of optimisation, so it is an open research area to create intelligent algorithms using RL. It is also a major challenge to generate a feasible initial solution for UETP using RL, as the final result entirely depends upon the initial solution. Use of deep RL provides an extra ability to explore the most optimised historical solution and generalise it for a new set of constraints to generate an optimised examination timetable.

Both UETP and UCTP are the most important managerial problems for universities and are performed at the same time, whereas classes of courses and examinations of different courses are executed in parallel due to the great number of semesters and different start and end timings of these semesters. Common resources such as rooms, teachers or invigilators are assigned for both of the activities. Therefore, proposing a solution for both problems at the same time is one of the most important and open research areas. Extending any developed and efficient algorithm to optimise with another problem will also be an important contribution. For this purpose, use of AI techniques with hybridisation are the most prominent requirement in developing such an algorithm.

CONCLUSIONS

This SLR provides a focused insight to explore various heuristic and meta-heuristic strategies implemented in the last five years to optimise UETP. The review aim is to provide the latest techniques and development to improve the performance and efficiency of optimisation algorithms by using AI and hybridisation. Thirty-two papers have been selected after following an extensive search policy in this paper and performing a detailed text review of all those papers. Clear-cut recommendations have been made to start new research in the field of UETP optimisation.

A variety of popular heuristic and meta-heuristic techniques are still used to solve problems, but they are not efficient in solving the problem in a way that satisfies all stakeholders. Even though they are not better in providing an efficient and effective solution, they can provide a better solution when combined in any particular way to develop a hybrid solution. Further improvement can be observed by implementing AI techniques in developing more robust and quick solutions for the problem, and its usage improves the performance of various algorithms or operations required to perform these techniques.

On review, it is also observed that the level of hybridisation can improve the results even more but for this purpose the use of AI can play a significant role. At the same time, solving examination scheduling with AI is still in infancy stage, so there is a huge gap to initiate research in this field to achieve the right mix of hybridisation as well as implementing AI to improve the performance of various algorithms. It provides a number of recommendations to extend or develop an optimisation strategy, use of hybridisation to improve the performance, and use of AI to get a smarter solution for the UETP.

AUTHOR CONTRIBUTIONS

Abdul Ghaffar: Conceptualization; investigation; writing – original draft; methodology; software; validation; formal analysis. **Irfan Ud Din:** Investigation; visualization; writing – review and editing; formal analysis; project administration; supervision; resources. **Asadullah Tariq:** Investigation; validation; visualization; writing – review and editing; data curation; formal analysis. **Mohammad Haseeb Zafar:** Funding acquisition; writing – review and editing; visualization; formal analysis; data curation; validation.

CONFLICT OF INTEREST STATEMENT

The authors affirm that they do not have any conflicts of interest to disclose regarding the current study.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

ETHICS STATEMENT

This article does not contain any studies involving human participants performed by any of the authors.

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